SECJ2203: Software Engineering

System Documentation (SD)

KADA Cooperative System

Version 1.0

Date

Faculty of Computing

Prepared by: KodeLab

Revision Page

a. **Overview**

This version 1.0 of the documents outlines how the system is designed based-on the requirements for the system. This includes the following section:

1. Introduction including purpose, scope, definition, acronym and abbreviations, references, and overview.
2. Specific requirements including user characteristics, system features, use case details, performance and other requirements, and design constraints.
3. Architectural rationale including architectural style and rationale.
4. Architectural views including use case view, implementation view, logical view, process view, and physical/ deployment view.
5. Data design including data dictionary for each entity.
6. User interface design including screen image.
7. Traceability of the system.

This version of the document acts as a blueprint for the stakeholders of the KADA Cooperative, future developer, and team members to understand and implement the system as the requirements needed.

b. **Target Audience**

- Stakeholder
- Development Team
- KADA Staff, Admin, and Board of Director

c. **Project Team Members**

The table below shows the list of the team members by stating their roles and status for each assigned task e.g. by section for this SD version (complete, partially complete, incomplete). If the assigned tasks are not done and have been assigned to other team members, state accordingly.

Member Name	Role	Task	Status
Lubna Al Haani Binti Radzuan (A23CS0107)	Team Leader	2.1 User Characteristic 2.3.3 Apply For Loan 2.3.9 Review Application 4.4 Process View 5.0 Data Design	Complete Complete Complete Complete Complete
Nabil Aflah Boo Binti Mohd Yosuf Boo Yong Chong (A23CS0252)	Team Member	2.3.1 Authentication 2.3.2 Apply For Membership 2.5 Design Constraints 7.0 Traceability	Complete Complete Complete Complete
Nur Firzana Binti Badrus Hisham (A23CS0156)	Team Member	2.2 System Feature 2.3.8 Manage Member 4.2 Implementation View	Complete Complete Complete

		6.0 User Interface Design	Complete
Nurul Adriana Binti Kamal Jefri (A23CS0258)	Team Member	1.0 Introduction 2.3.6 View Financial Report 2.3.7 BOD Report 4.3 Logical View 4.5 Physical/ Deployment View	Complete Complete Complete Complete Complete
Pravinraj A/L Sivabathi (A23CS0171)	Team Member	2.3.4 View Financial Statement 2.3.5 Member Statement 2.4 Performance and Other Requirement 3.0 Architectural Rationale 4.1 Use Case View	Complete Complete Complete Complete Complete

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Team leader (Lubna Al Haani) Member (Nabil Aflah Boo, Nur Firzana, Nurul Adriana, Pravinraj)	Completed Section 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, and 7.0.	21/12/2024

Note:
This System Documentation (SD) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998), Software Design Descriptions (SDD) (IEEE Std. 10161998 1), and Software Test Documentation (IEEE Std. 829-2008) that are simplified and customized to meet the need of SECJ2203 course at Faculty of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly.

Table of Contents

1. Introduction

This System Documentation (SD) has detailed explanations related to the components and processes involved in designing the KADA Cooperative System that include System Requirements Specification (SRS), System Design Document (SDD), and System Testing Document. This paper, therefore, gives clarity that can be understood by the stakeholders or members of the team.

1.1 Purpose

The SD indicates a design and strategy for implementation of the KADA Cooperative System, an e-cooperative that intends to digitize the cooperative operations. It clearly specifies the technical and functional specification required for development, testing, and deployment.

The target audience includes:

- Development Team Members: To know the technology and validate the system's requirements.
- KADA Company: To verify that the system goes with cooperation goals and operational requirements.
- Lecturers: In charge of operating and analyzing the progress and compliance of the project.

1.2 Scope

The KADA Cooperative System is aimed at enhancing the core functionalities of the cooperative by developing an automated and contemporary platform in replacement of the existing manual paper-based process. Important features include the following:

a) Online applications for membership and loans.
b) Real-time reporting and membership management.
c) Provide a secure role-based access to applicants, members, administrators and board members.

This system is meant to boost the operational efficiency, user experience and integrity of the data. It will not include irrelevant items, such as those which have a third party contact with non-cooperative financial services.

1.3 **Definitions, Acronyms and Abbreviation**

Term	**Definition**
SRS	System Requirements Specification
SDD	System Design Document
SD	System Documentation
KADA	Kemubu Agricultural Development Authority
BOD	Board Of Directors
Admin	System Administrator
SSD	Solid-State Drive

1.4 **References**

[1] Dennis, A., Wixom, B. H., & Tegarden, D. (2015). *Systems Analysis and Design: An Object-Oriented Approach with UML* (A. Dennis, B. H. Wixom, & D. Tegarden, Eds.). Wiley.

1.5 **Overview**

The SD is divided into the following sections:

- Specific Requirements: Outline the system characters, characteristics, and limitations.
- Design Components: These include use-case diagrams, class diagrams, and state diagrams.
- Launch Phase: Definition of sprints and user stories for iterative development.
- Testing and Validation: Handling of requirements from both the link in functional and non-functional attributes, performance measures, and quality standards.

This systematic arrangement assures that the document serves as an all-around reference material for all phases of the project, thus allowing for effectual collaboration and successful submitting of the whole KADA Cooperative System.

2. Specific Requirements

This section describes the requirements of the system functional and non-functional, including the user characteristics, system features, and use case details for each use case.

2.1 User characteristics

This section describes the characteristics of the users who will interact with the KADA Cooperative System. This is to provide the development team members with clear understanding and to ensure the design of this system is aligned with the user's needs, abilities and expectations.

There are three main types of users that will be using this system: KADA staff, KADA Cooperative admin and board of directors (Ahli Lembaga Koperasi).

2.1.1 KADA Staff

- The KADA staff who will use the system are expected to have access to devices for web-based systems.
- They may have varying levels of computer skills and may have difficulty understanding how to use the web-based systems.
- Training or user manual can be provided if needed.
- They will use the system to fill-in an online form to register as cooperative members and also apply for a loan if wanted.
- They also can use the system to be updated

2.1.2 KADA Cooperative Admin

- The KADA cooperative admin who will use this system is expected to have intermediate to advanced computer skills including proficiency in using web-based platforms, office productivity tools and familiarity with database systems for managing large amounts of data.

1. Manage and monitor data
    - The KADA cooperative admin will use the system to monitor all records of KADA staff that register into the system.
    - They also will update and maintain accurate financial member data in the system.

2. Financial management
    - Manage and monitor financial statements for KADA members monthly.
    - Generate and manage financial reports monthly and annually for KADA Cooperative’s board of directors.

2.1.3 KADA Cooperative Boards of directors

- KADA Cooperative board of directors who will use this system are expected to have basic to intermediate computer skills including the familiarity with viewing reports and dashboards.
- KADA Cooperative board of directors is also expected to have limited technical expertise, making simplified and user-friendly interfaces helpful for them.
- KADA Cooperative board of directors will use the system to review financial reports of KADA Cooperative.
- They also will use the system to review member registration and loan applications from KADA staff.

2.2 System Features

This product is a new website system for KADA Cooperative which is aimed at making a complete change from the hard copy methods of processing membership and loan applications to more automated systems. It tries to help the user eliminate all the inconveniences that are associated with applying for membership and loans, managing user details and even managing reports. In addition, the system incorporates the capabilities of calculating cooperatives' loans and interests for each member, reducing the workload on the people.

The system features are illustrated in Figure 2.2.1 below. The detailed description of each module and functions is tabulated in Table 2.2.1.

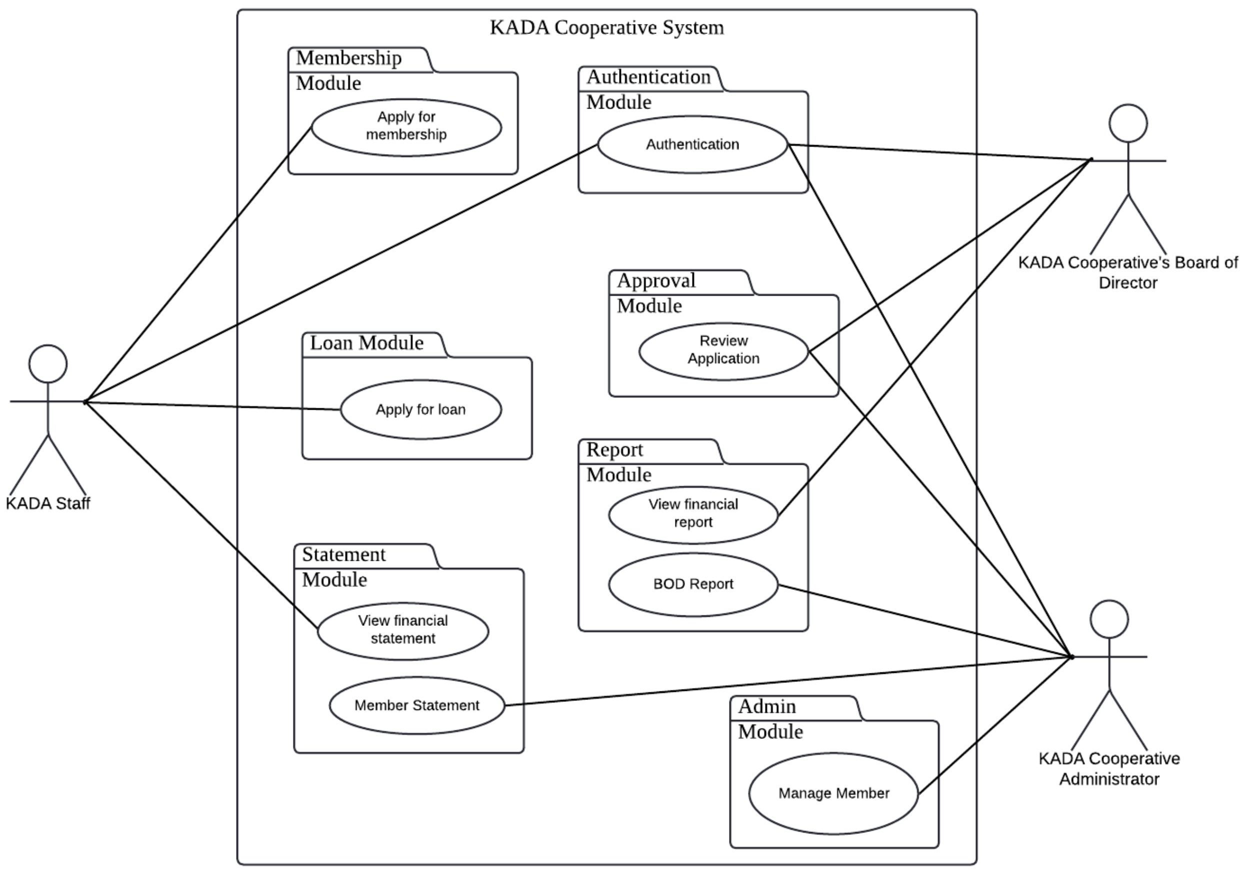


Figure 2.2.1: Use Case Diagram for KADA Cooperative System

Table 2.2.1: Description of Module and Functions for KADA Cooperative System

Module	Function	Description
Authentication	UC001-Authentication	This use case allows all users including staff, admin and BOD to login to the Cooperative system.
Membership	UC002-Member Registration	This use case allows non-member staff to register membership in KADA Cooperative.
Loan	UC003-Loan Application	This use case allows members to submit a loan application along with necessary documents.
Statement	UC004-View Financial Statement	This use case allows both members and admin to view member-specific financial details, including balances and transaction histories.
	UC005-Member Statement	This use case allows members to view their statements.
Reporting	UC006-View Financial Report	This use case allows administrators or Board of Directors (BOD) members to view comprehensive financial reports for the organization.
	UC007-BOD Report	This use case provides detailed reports specifically designed for Board of Directors' decision-making purposes.
Admin	UC008-Manage member	This use case allows system administrators to manage user roles, permissions, and system settings.
Approval	UC009-Review Application	This use case allows BODs to decide the membership and loan applications.

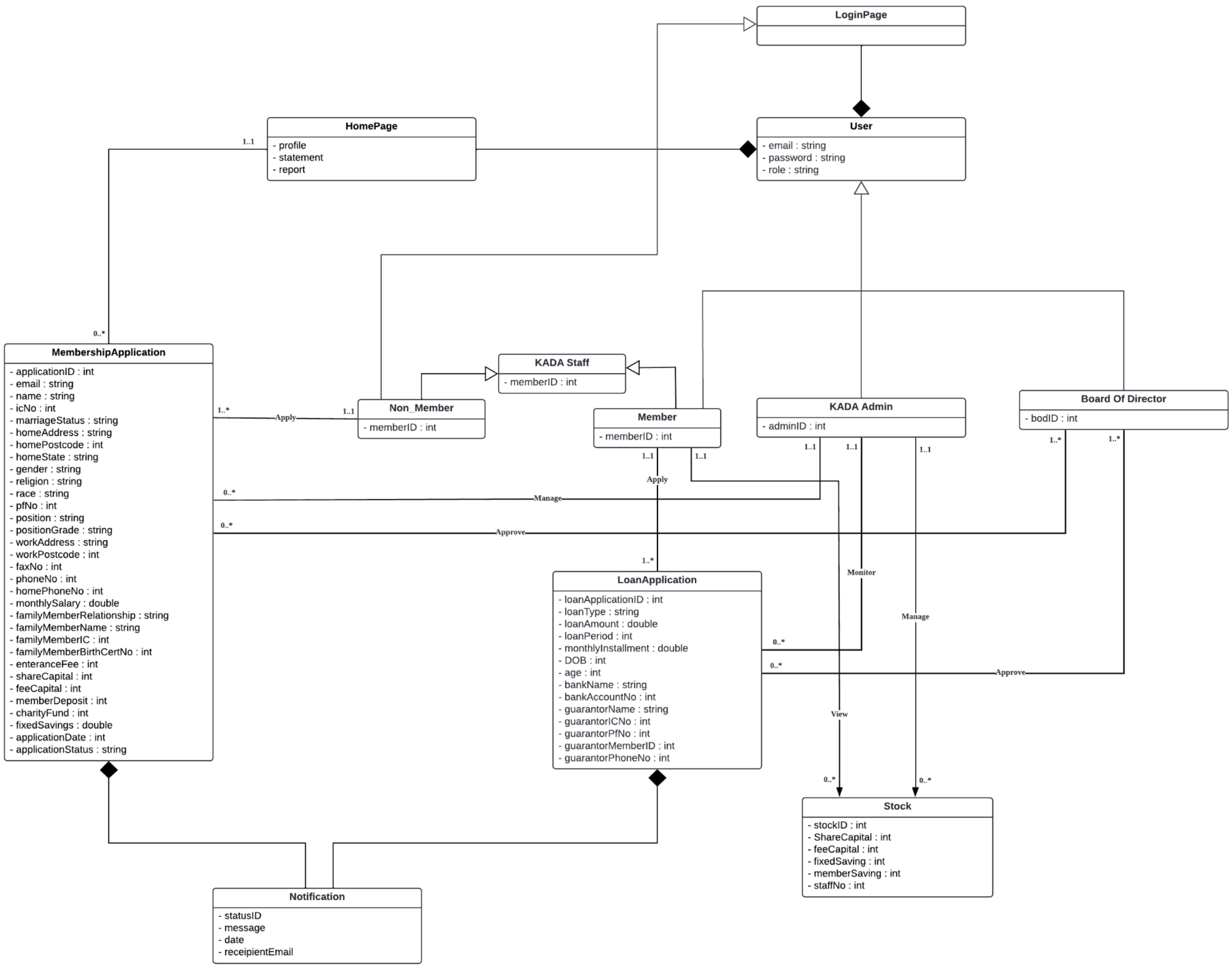


Figure 2.2.2 : Domain Model for KADA Cooperative System

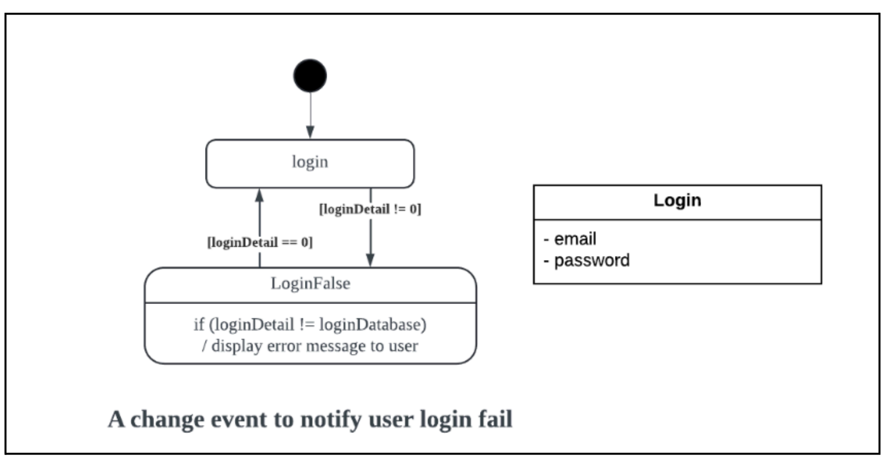


Figure 2.2.3 : State Diagram for authentication and authorization

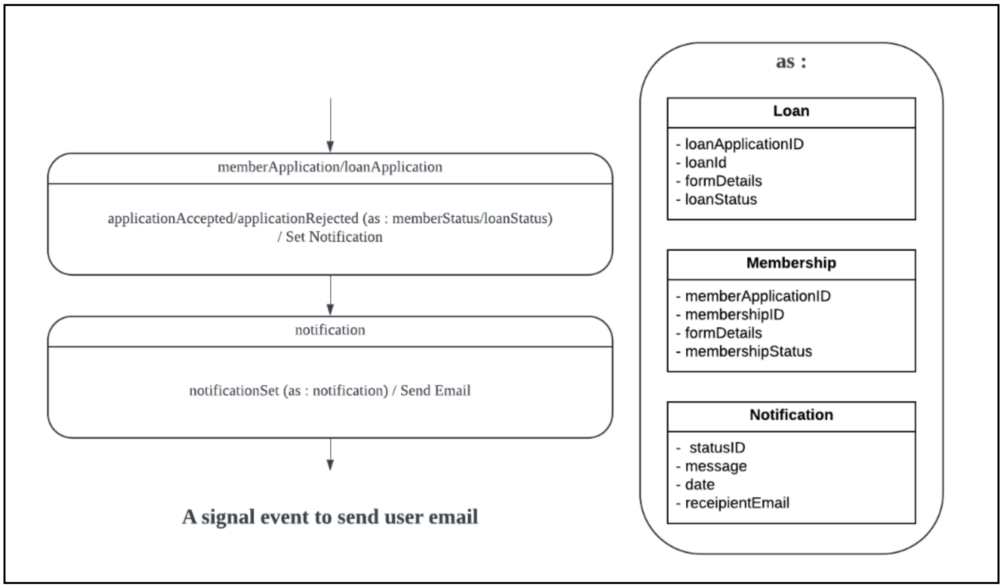


Figure 2.2.3 : State Diagram for Loan Application

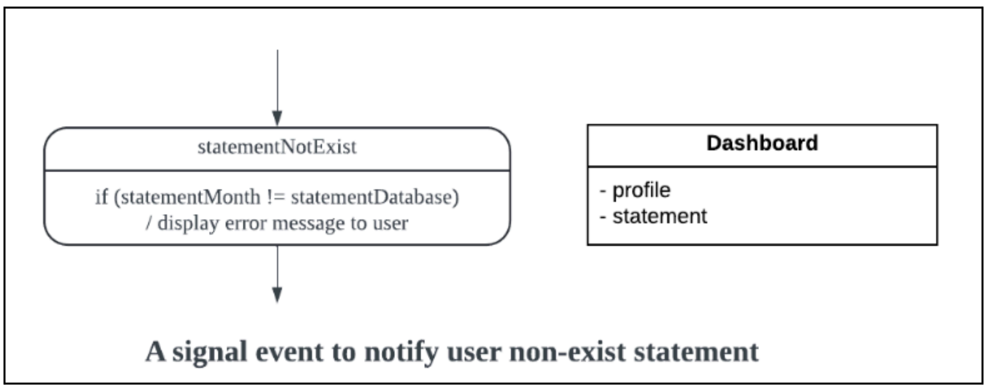


Figure 2.5 : State Diagram for Dashboard

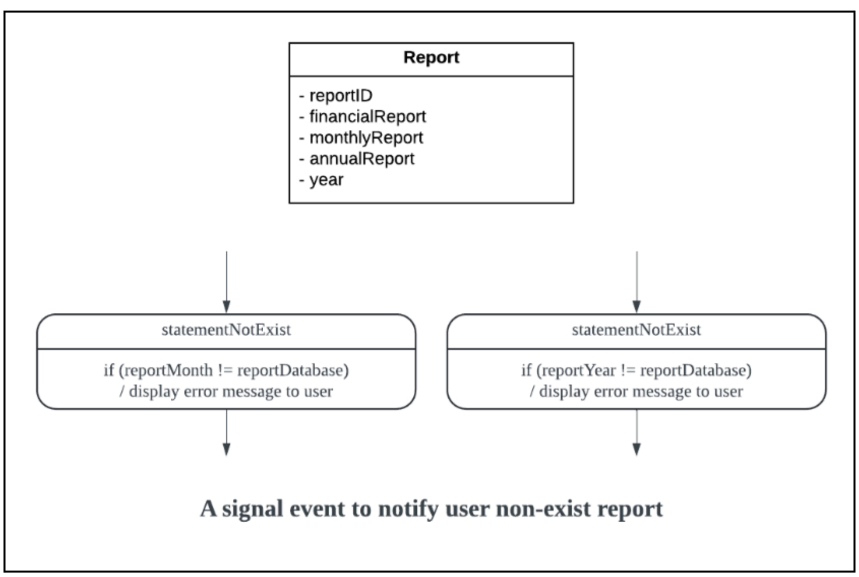


Figure 2.6 : State Diagram for Report

2.3 Use Case Details

This section provides detailed descriptions of each use case for the system. It outlines the interactions between the actor and the system by showing the activity diagram and sequence diagram of the system.

2.3.1 UC001 : Use Case <Authentication>

Table 2.3.1.1: Use Case Description for Authentication

Use case: Authentication
ID: UC001
Actors: KADA Staff, KADA Admin, KADA BOD
Preconditions: 1. The actors must have the valid email and password to login the system. 2. The actors must have a stable internet connection to be able to enter the system.
Flow of events: 1. The user clicks onto the login page of the system. 2. The system prompts the user the login form. 3. The users enter their email address and password through the login page. 4. The system verified the user authentication. 5. If the authentication is valid, the system grants the user access to the system based on their roles. 6. If the authentication is invalid, the system prompts the error message and denies the user access to the system.
Alternative flow 1: If the user enters an invalid email address or password, the system prompts an error message and requires the user to enter the valid email and password.
Alternative flow 2: If the user has not registered to the system, they need to click on the sign up page and enter their email address and create a new password.
Postconditions: Upon the valid authentication process, the user will be redirected to the user interface according to their roles in the system.

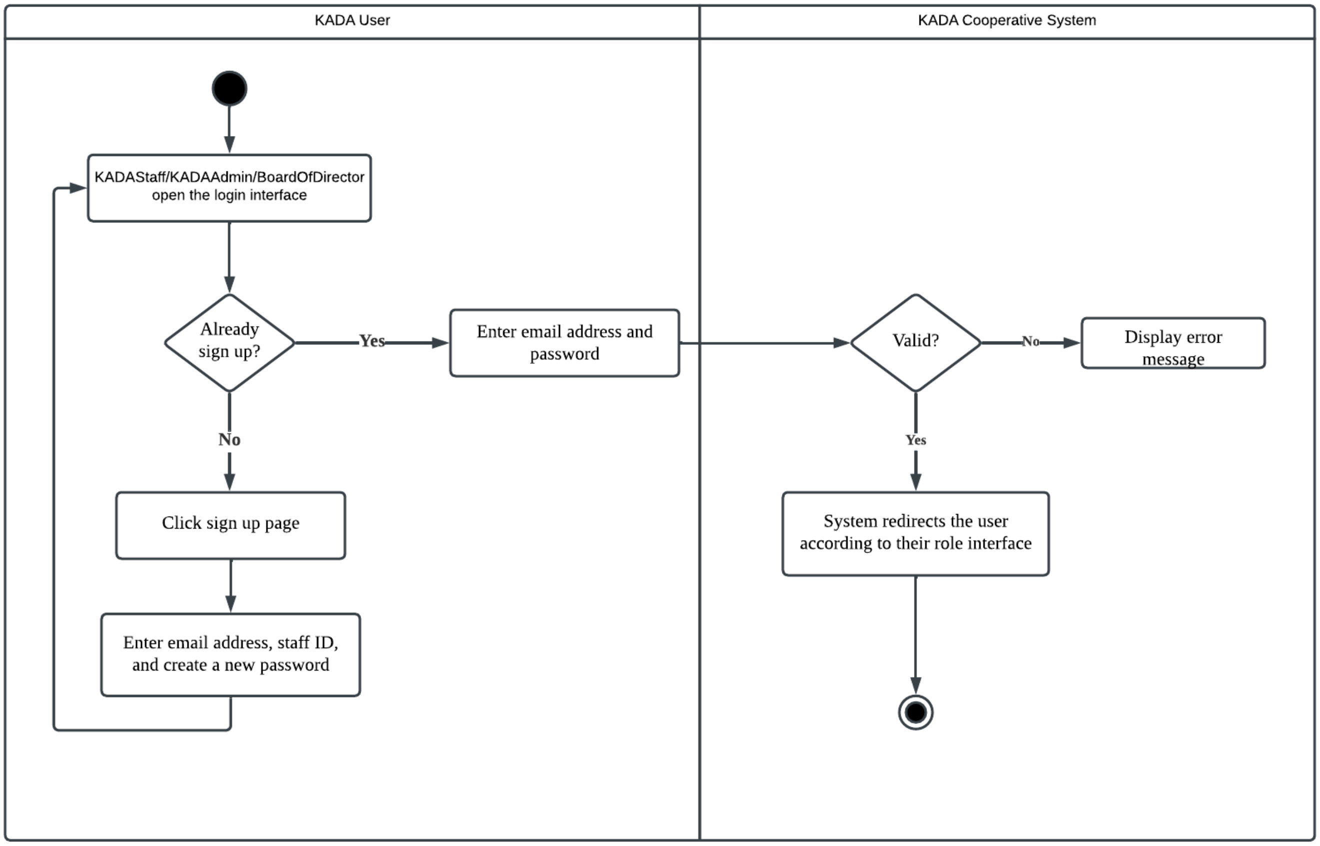


Figure 2.3.1.2: Activity Diagram for Authentication

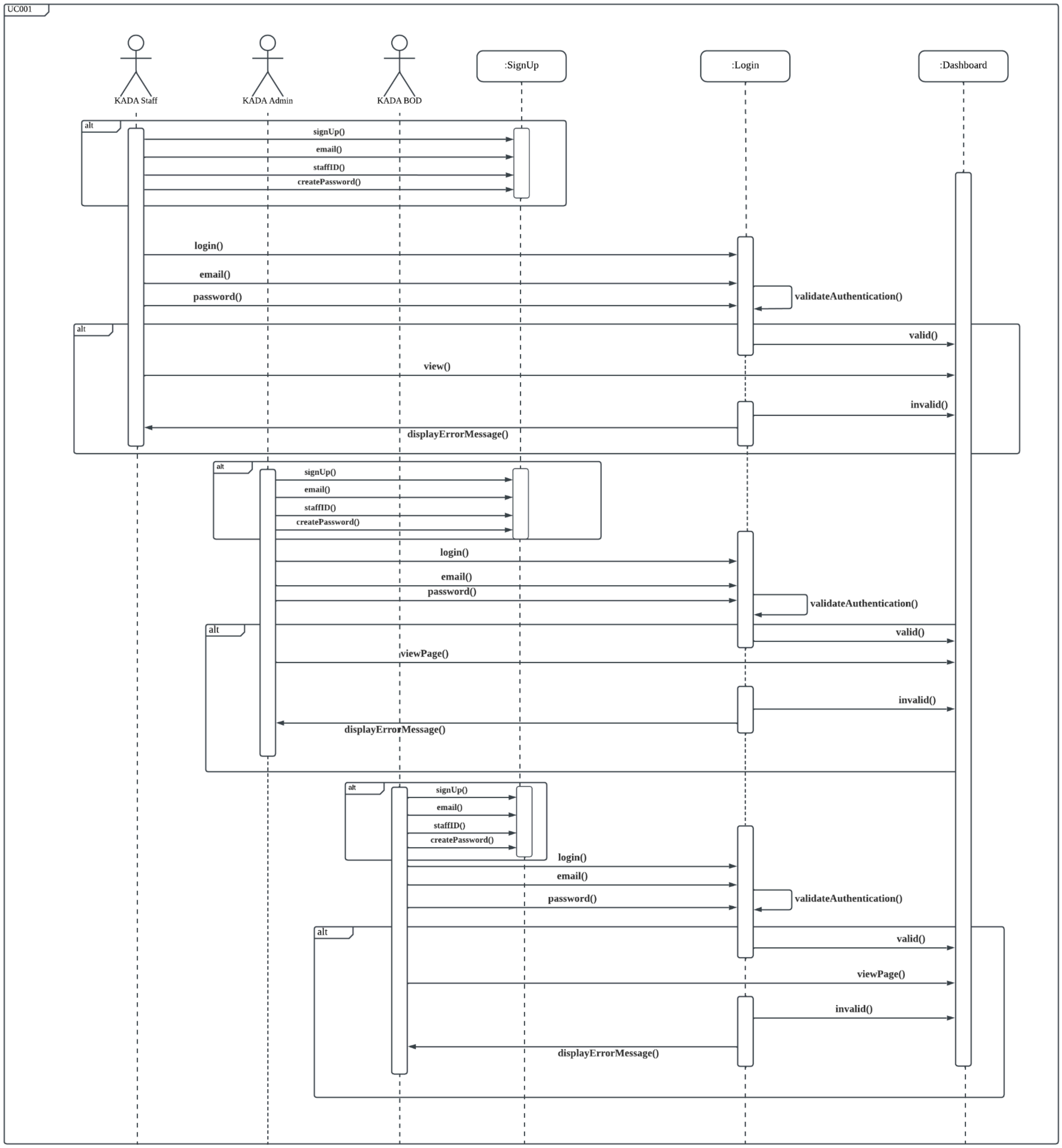


Figure 2.3.1.1: Sequence Diagram for Authentication

2.3.2 **UC002: Use Case <Apply For Membership>**

Table 2.3.2.1: Use Case Description for Apply For Membership

Use case: Apply For Membership
ID: UC002
Actors: KADA Staff
Preconditions: 1. The Staff must have the valid email and password to login the system. 2. The Staff that apply for the membership should be the one who is not a member yet.
Flow of events: 1. The Staff logs into the system. 2. The Staff navigate themselves to the “Permohonan Ahli” page. 3. The Staff required to fill all the information given and membership declaration. 4. The Staff clicks on the “hantar” button to submit the application. 5. The system prompts the successful submitted message.
Postconditions: 1. Once the application has been submitted, the Staff cannot make any changes to the application. 2. The Staff can track their membership application status after submission at the “Permohonan Ahli” page.

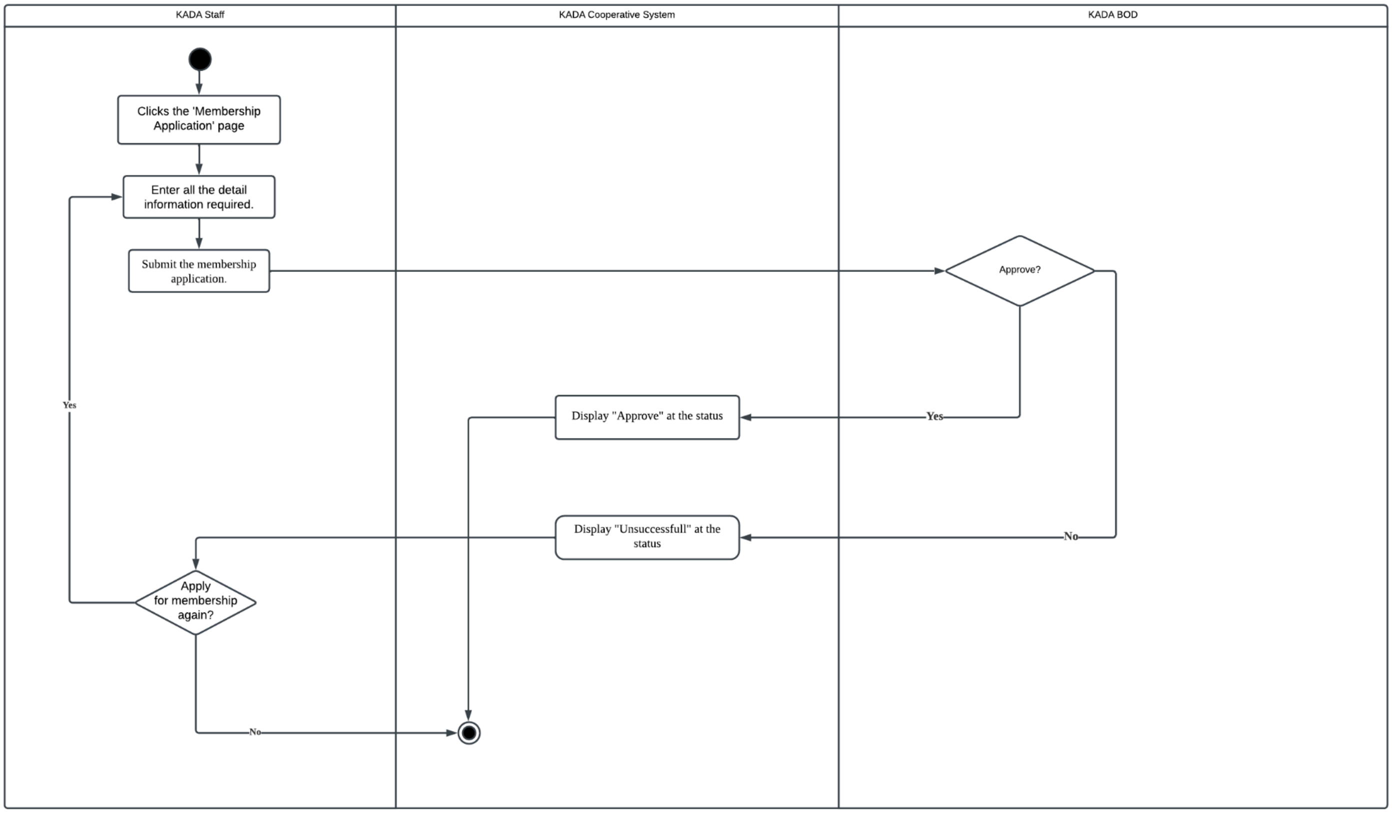


Figure 2.3.2.2: Activity Diagram for Apply For Membership

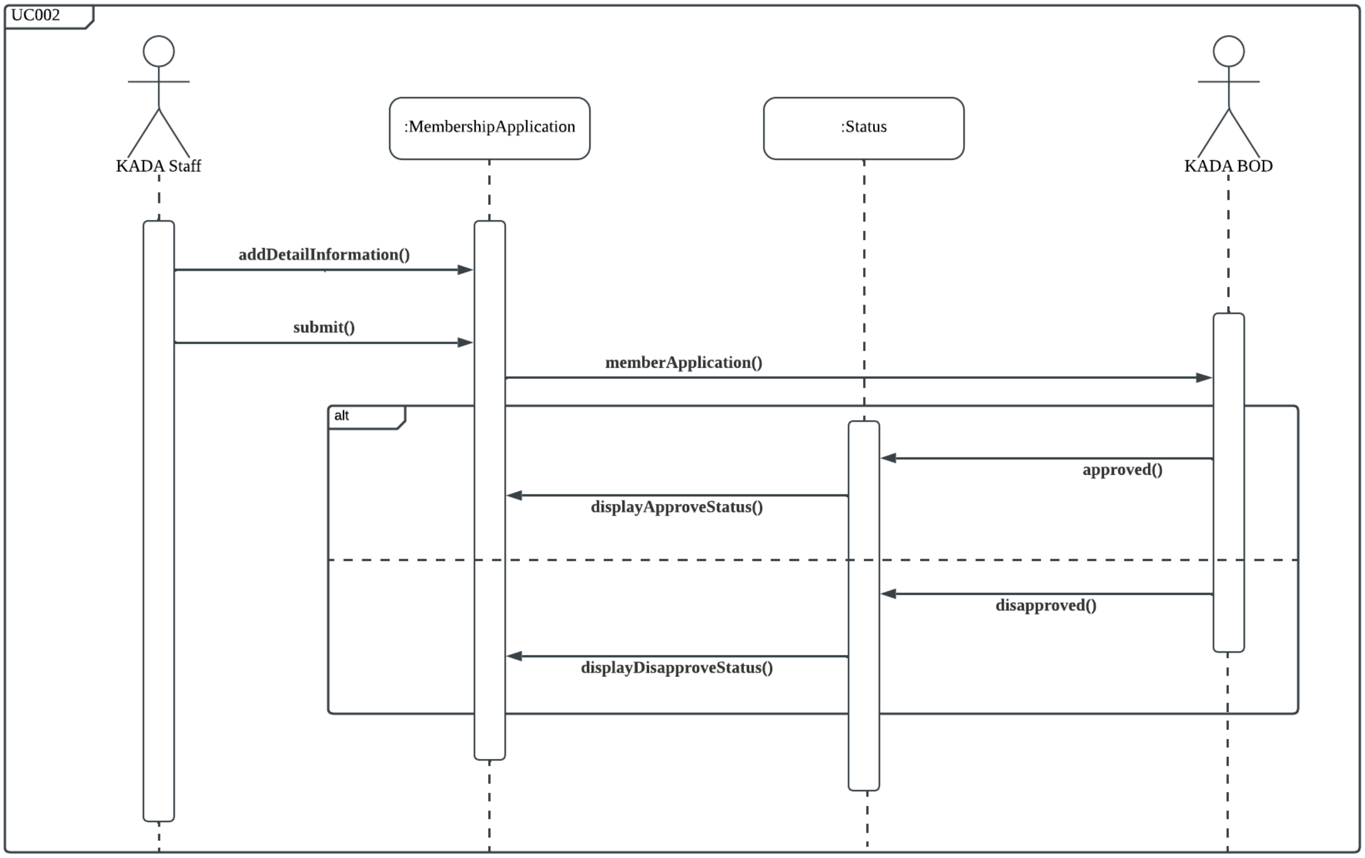


Figure 2.3.2.1: Sequence Diagram for Apply For Membership

2.3.3 **UC003: Use Case <Apply for Loan>**

Table 2.3.3.1: Use Case Description for Apply for Loan

Use case: Apply for Loan
ID: UC003
Actors: KADA Staff
Preconditions: 1. The staff that apply for loan should be a member and have paid completely the modal share to the cooperative.
Flow of events: 1. The KADA staff will select “Permohonan Pembiayaan” on the homepage. 1.1. System will display the loan details page. 2. KADA staff will select “Mohon Pembiayaan” button. 2.1. System will display the loan application page. 3. KADA staff will fill-in all details for the loan application. 4. KADA staff select the “Hantar” button. 4.1. System will validate all details. 4.2. If all details are valid, the system will send data into the system to be reviewed. 4.3. Else, the system will display an error message.
Postconditions: 1. After the application has been reviewed, KADA staff can view application status via application status page. 2. KADA staff select “Status Permohonan”. 2.1. If loan application exist 2.1.1. The system displays loan applications with status. 2.2. Else 2.2.1. The system will display an error message.

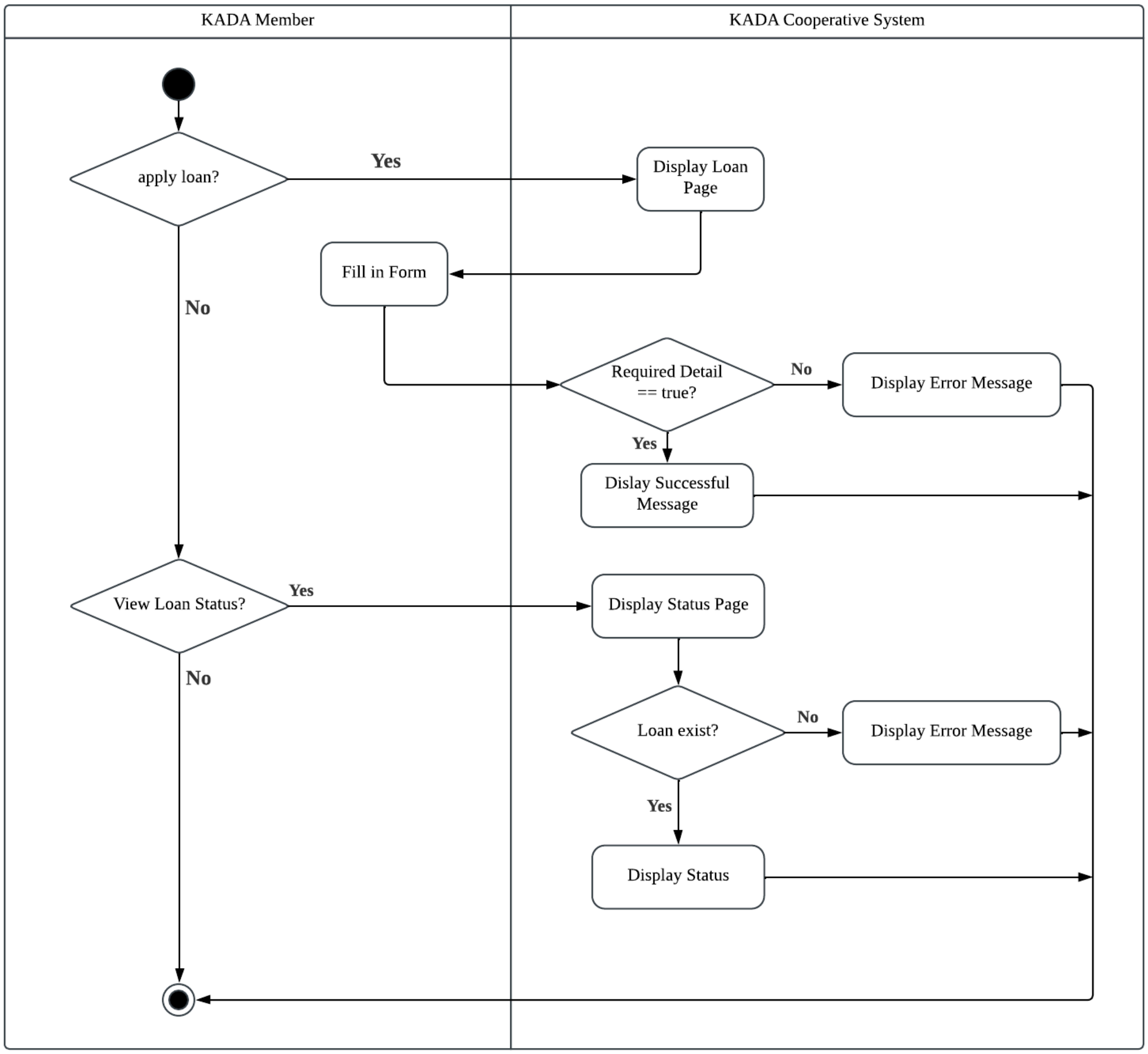


Figure 2.3.3.1: Activity diagram for Apply for Loan

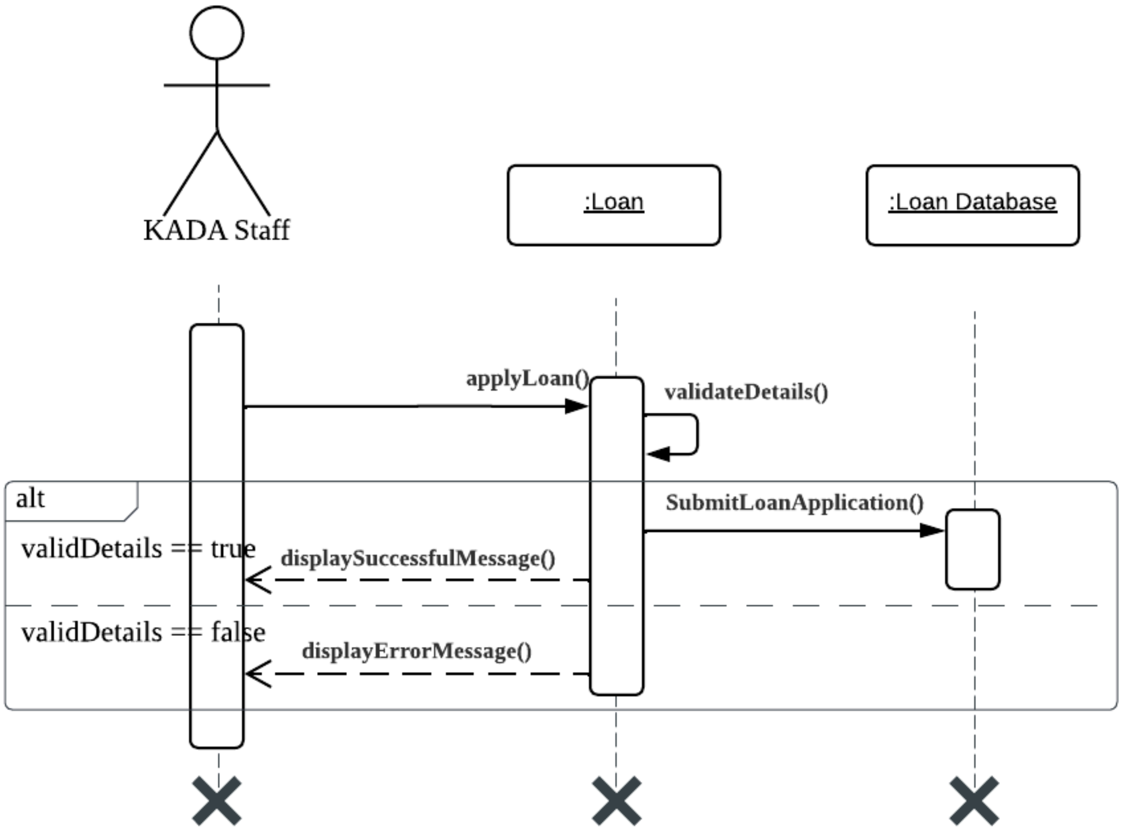


Figure 2.3.3.2: Sequence diagram for Apply for Loan

2.3.4 **UC004: Use Case <View Member Statements>**

Table 2.3.4.1: Use Case Description for View Member Statements

Use Case : Member Statement
ID: UC004
Actors: Kada Staff
Preconditions: 1. Kada Staff must be logged in to the KADA system to view member statements.
Flow of events: 1. Kada Staff logs into the KADA system 2. Kada Staff enters the 'View statement' section 3. Kada Staff selects the type of statement that they want view 4. If Kada Staff selects savings balance 4.1. System retrieves and displays the staff's history of transaction and savings balance amount. 5. If Kada Staff selects loan balance 5.1. System retrieves and display the staff's loan balance 6. Kada Staff can export the statement in desired format if needed
Postconditions: 1. Kada Staff views and saves or prints the financial statement

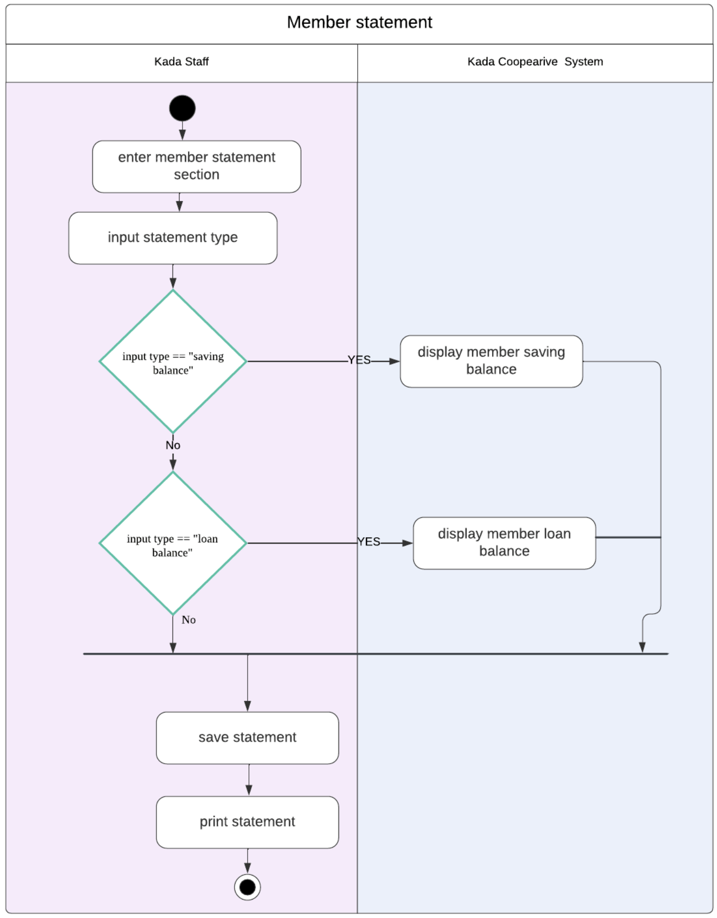


Figure 2.3.4.2: Activity diagram for View Financial Statement

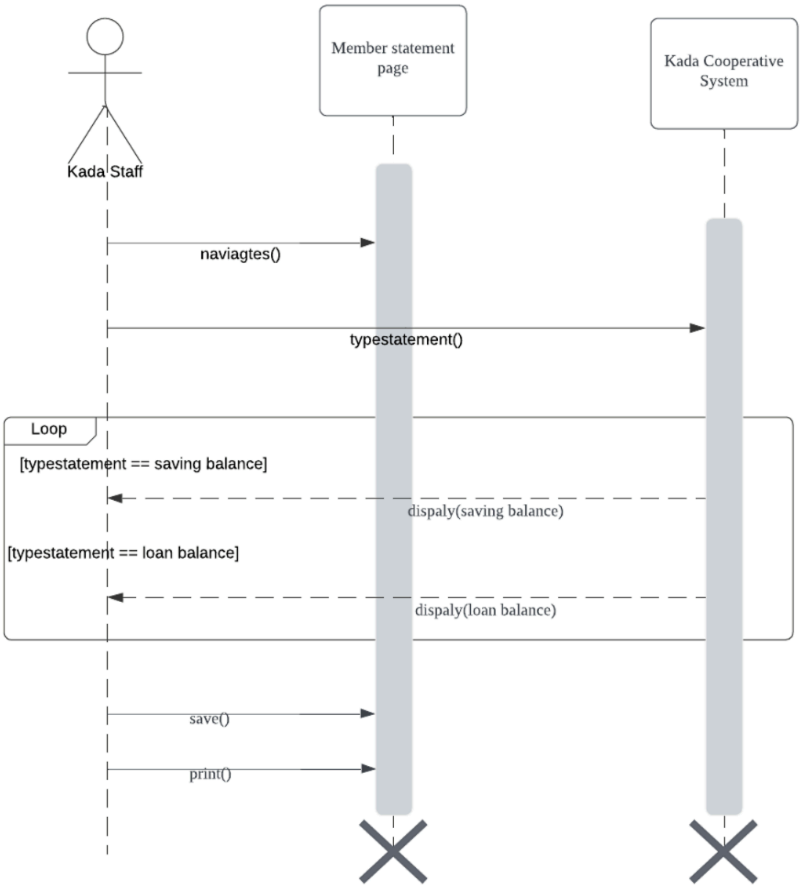
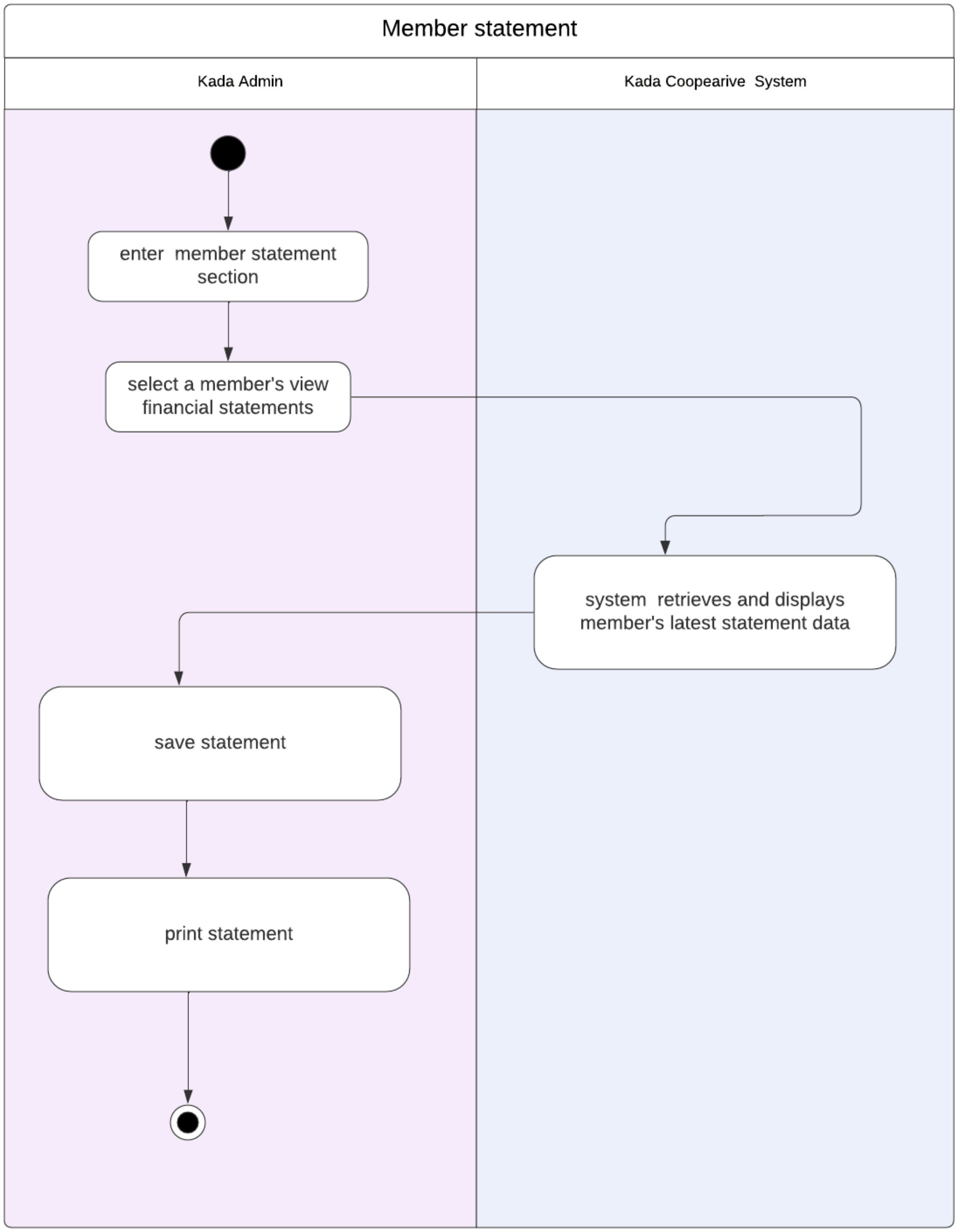


Figure 2.3.4.3: Sequence diagram for View Financial Statement

2.3.5 **UC005: Use Case <Member Statement>**

Table 2.3.5.1: Use Case Description for Member Statement

Use case : Member statement
ID : UC005
Actors : Kada Admin
Preconditions: 1. Kada Admin must be logged in to the KADA system to access member's financial statements.
Flow of events: 1. Kada Admin must log into the system 2. Kada Admin enter 'Member statement' section 3. Admin find and select specific member's view financial statements to view 4. System retrieves and displays member's latest statement data which are savings and loan balances 5. Kada Admin can save or print it for future use
Postcondition: 1. Kada Admin views and optionally saves the financial statement for administrative process



Fogure 2.3.5.2: Activity diagram Description for Member Statement

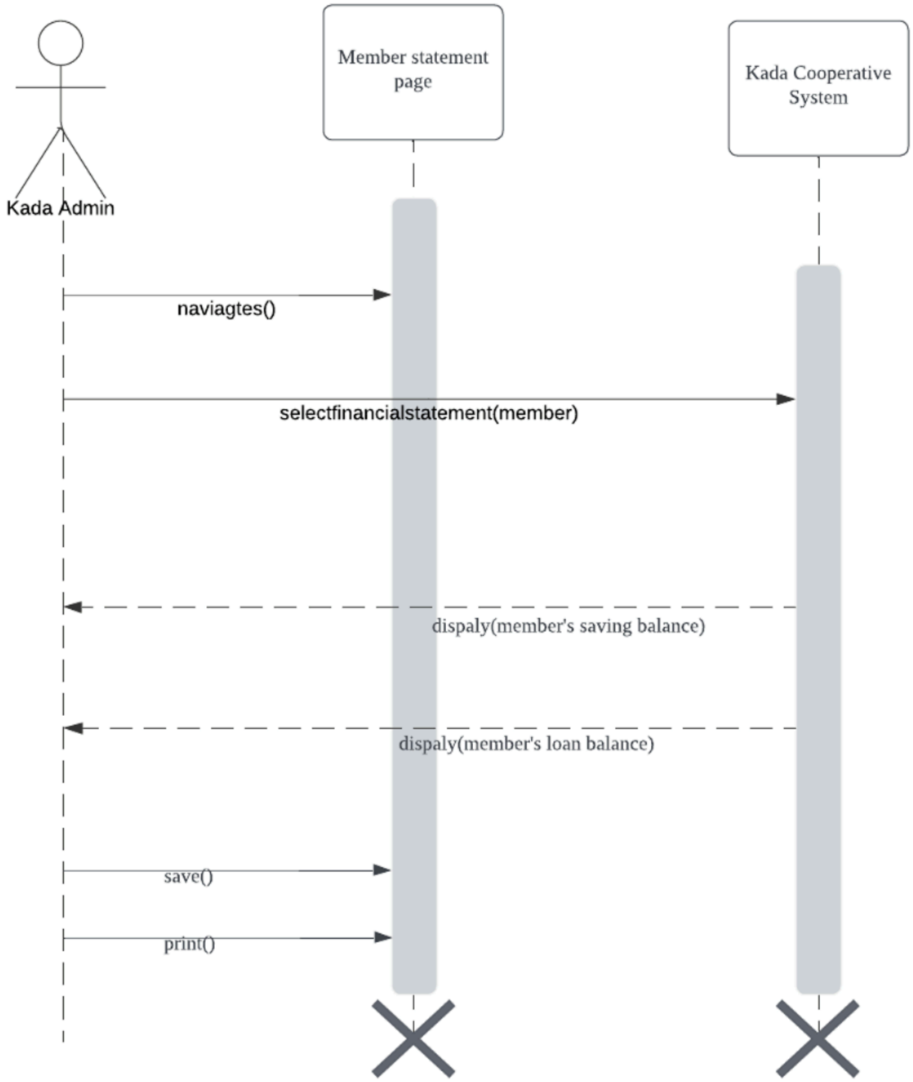


Figure 2.3.5.3: Sequence diagram Description for Member Statement

2.3.6 UC006 : Use Case <View Financial Report>

Table 2.3.6.1: Use Case Description for View Financial Report

Use Case : View Financial Report
ID : **UC006**
Actors: KADA Cooperative Board of Directors (BOD)
Preconditions: 1. The BOD member has to log into the KADA Cooperative System. 2. Finance-related data must be recorded and are available in the database.
Flow of events : 1. The KADA System directs the BOD member to the dashboard. 2. The BOD member navigates to the "Financial Report" section. 3. The KADA System displays a list of financial reports that can be accessed. 4. The BOD member selects a certain financial report type like monthly profit report or loan distribution report. 5. The KADA System obtains the required information and prepares an appropriate report. 6. The BOD member examines the report data on screen. 7. The BOD member may download the report in PDF or Excel format.

Alternative flow :
● If financial data is not available
    1. The KADA system will display a message saying "No data accessible for the chosen type of report or period".
    2. The BOD member can select another kind of report or exit the "Financial Report" section.
● If there's error in system
    1. It will process it in logs and also notify the member of the BOD: "An error has occurred in generating the report. Kindly try later."

Postcondition : 1. The BOD Member is now able to access the relevant financial statement from the KADA System quite successfully. 2. If it is needed, the BOD member can download the report for use or further analysis.

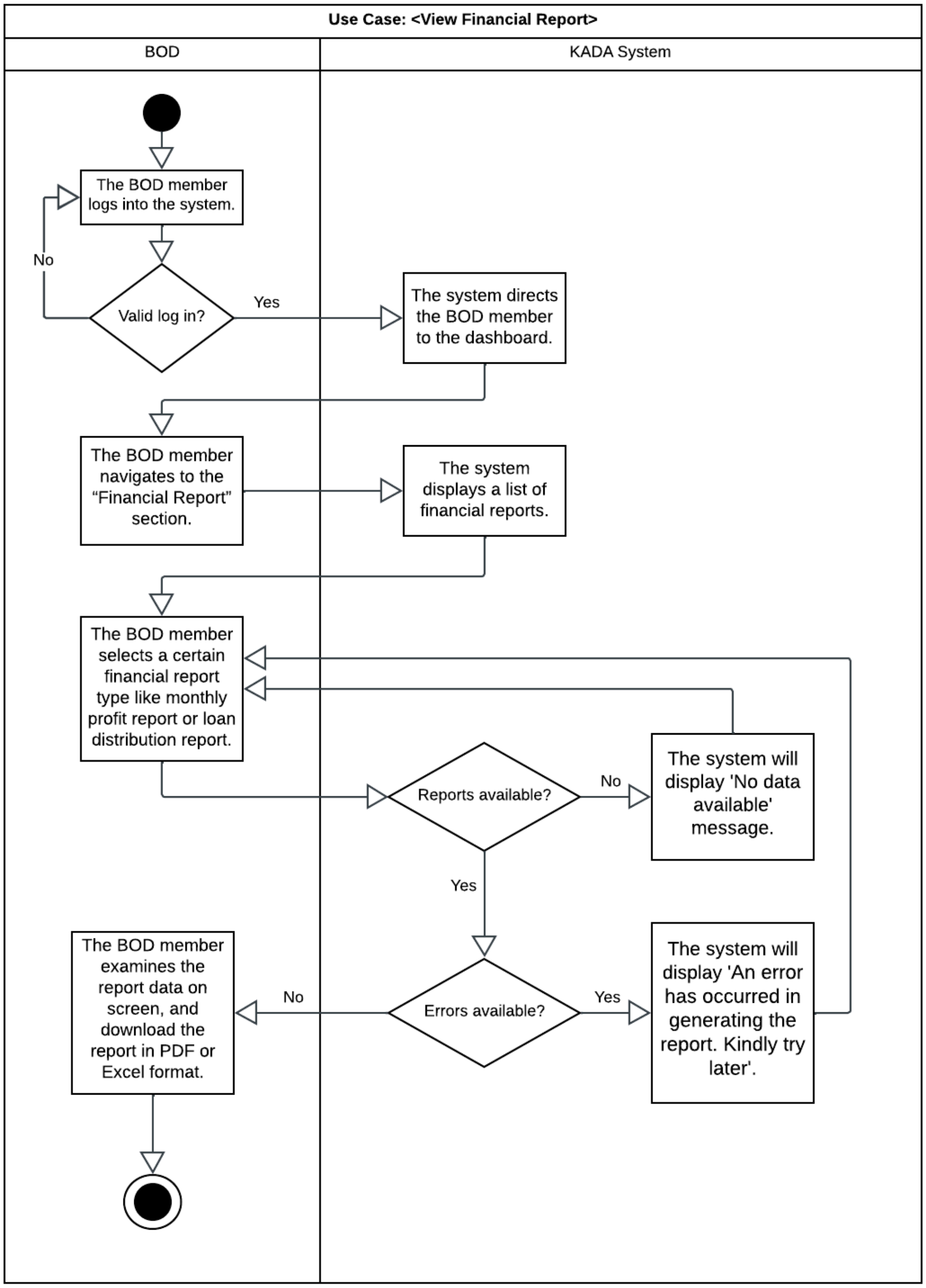


Figure 2.3.6.1: Activity Diagram for View Financial Report

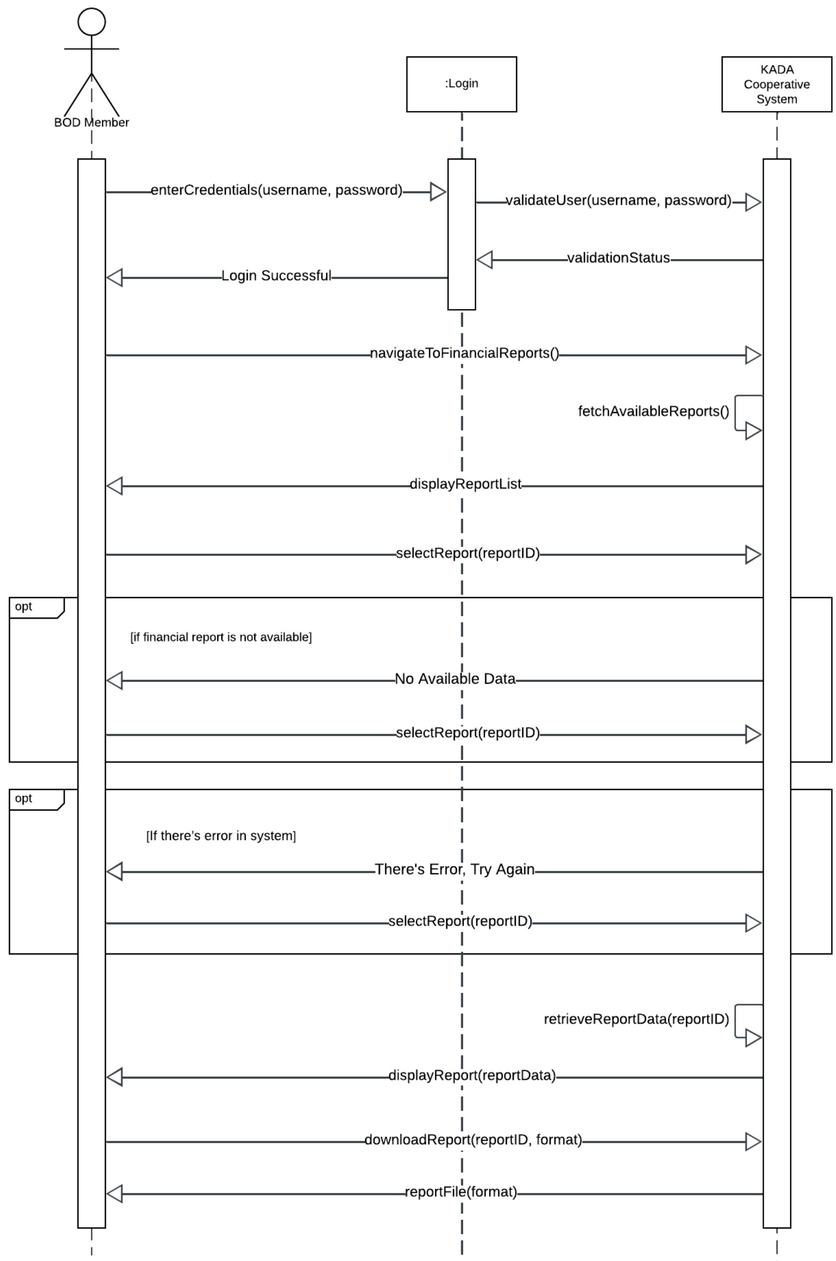


Figure 2.3.6.2: Sequence Diagram for View Financial Report

2.3.7 **UC007: Use Case <BOD Report>**

Table 2.3.7.1: Use Case Description for BOD Report

Use Case : BOD Report
ID : **UC007**
Actors: KADA Cooperative Board of Directors (BOD)
Preconditions: 1. The BOD member has to log into the KADA Cooperative System. 2. The report must be built upon existing financial and operational inputs within the system.
Flow of events : 1. The KADA System provides access. 2. The BOD member navigates to the "BOD Report" section. 3. The KADA System presents available options for creating different types of reports (membership, loan approvals, and financial summary). 4. The BOD member generates reports by selecting the appropriate report type and filtering data (date range, loan type, etc.). 5. The KADA System retrieves such information and displays the prepared report. 6. The BOD member can sort or dynamically filter that report. 7. The report will be available as a download in PDF or Excel format to the BOD member.

Alternative flow :
● If options entered is invalid
    1. The KADA system will throw an error stating "Invalid filter settings, please change your inputs”.
    2. The BOD member can alter the filters and try once again.
● If data is not available
    1. The KADA system displays the message that says: "No data is available based on the filters or the report type chosen."
    2. Either the BOD member could alter the filters or exit the section.

Postcondition : 1. The BOD member gets to successfully check the request report or down it. 2. The KADA System tracks report access for audit reasons.

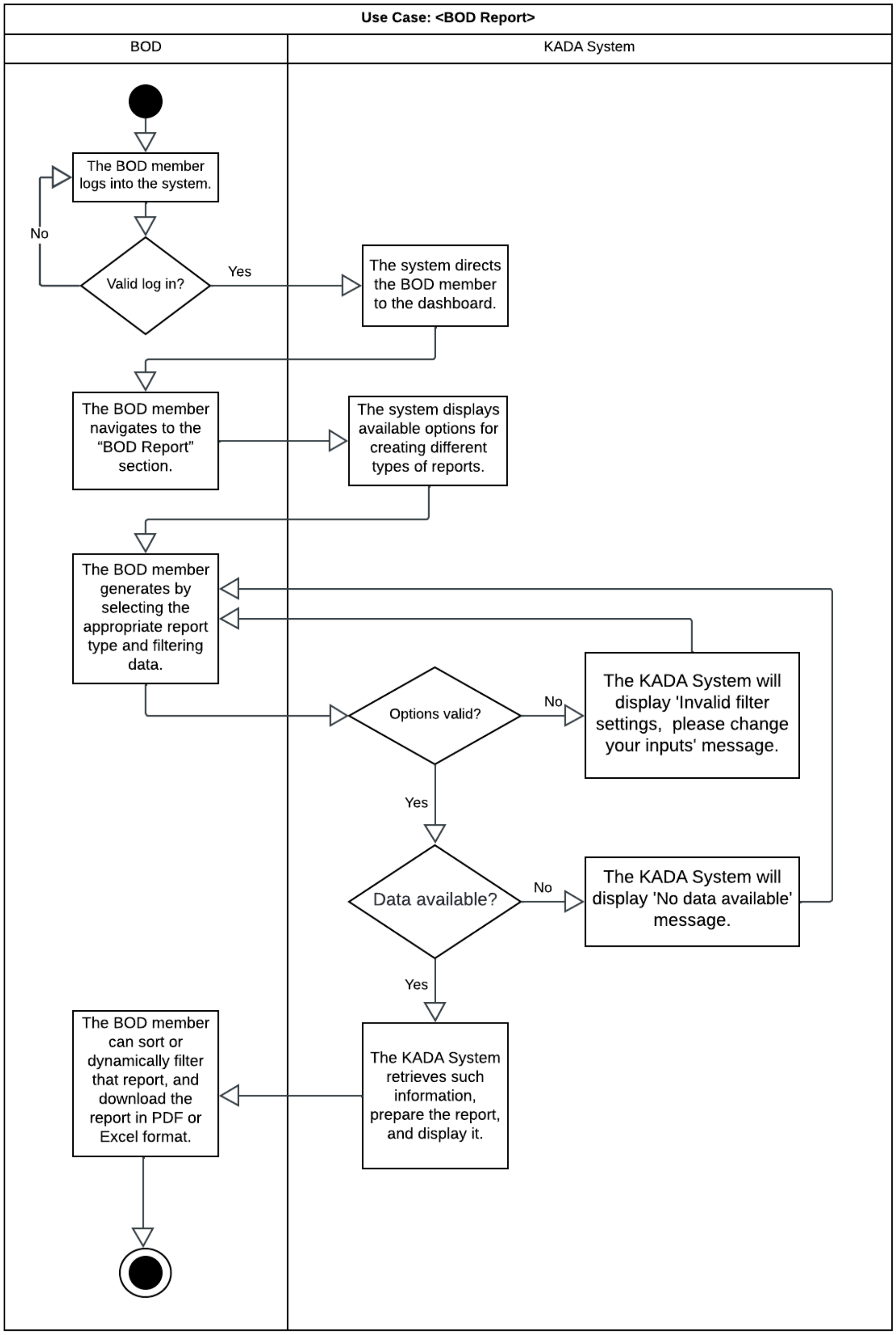


Figure 2.3.7.1: Activity Diagram for BOD Report

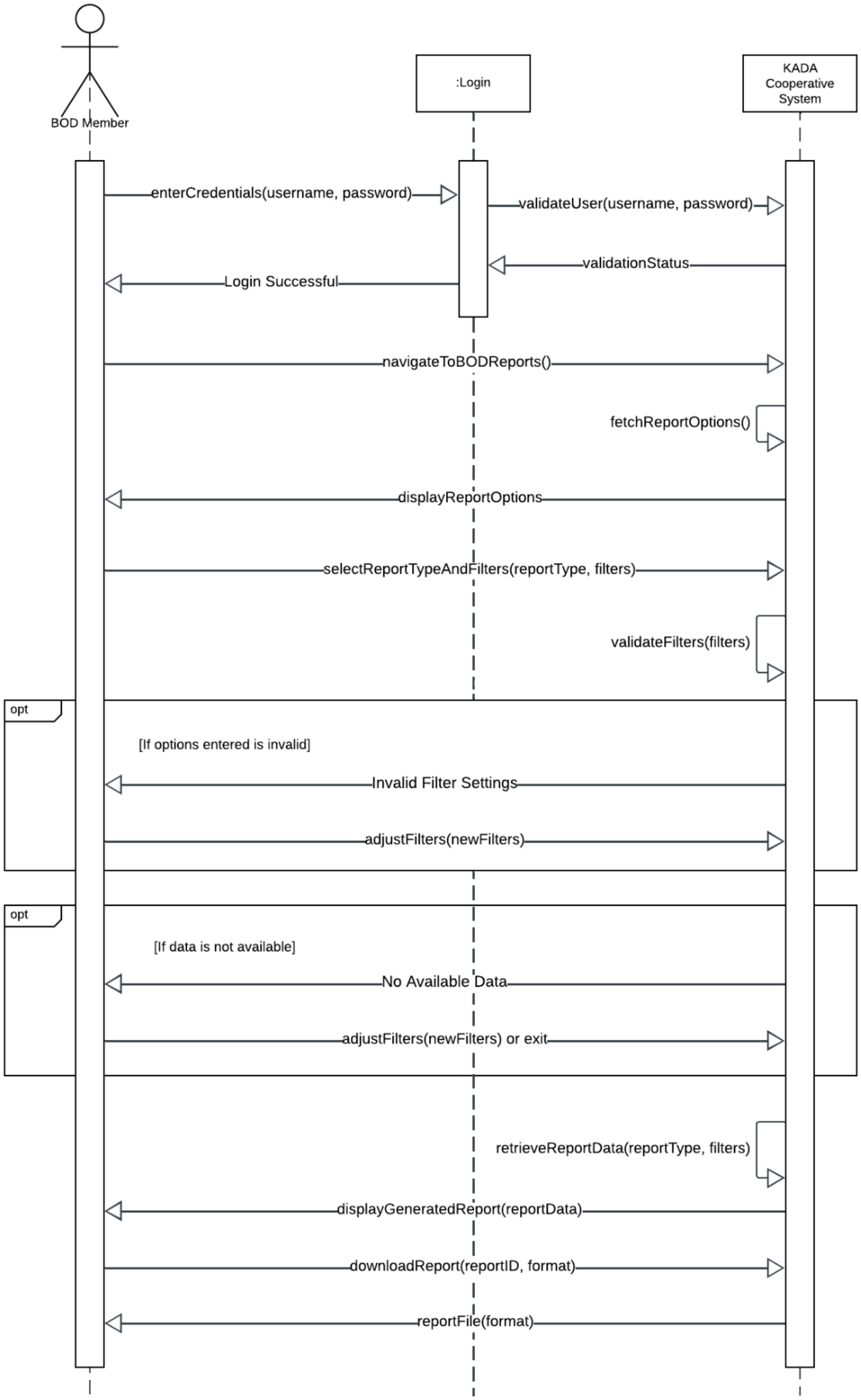


Figure 2.3.7.2: Sequence Diagram for BOD Report

2.3.8 **UC008: Use Case <Manage Member>**

Table 2.3.6.1: Use Case Description for Manage Member

Use Case : Manage Member
ID : **UC008**
Actors: KADA Admin
Preconditions: The user must be an authorized KADA Admin with appropriate system access.
Flow of events : 1. Admin logs into KADA System with their specified email and password. 2. System grants access to the admin page, providing options for editing and monitoring the system, managing the data and user management. 3. Admin views the details of the applicants of the membership and loan applications. 4. Admin manages the application status and system sends the notifications via email to the applicants. 5. Admin selects a specific task such as editing the home page, managing the application statuses, editing the statement or updating the reports. 6. System performs the requested task and confirms the successful completion.
Alternative flow : Unauthorized Access : If an unauthorized user attempts to access the administration dashboard, the system denies the access and logs the attempt.
Postcondition : Admin tasks are performed successfully to ensure the system integrity and availability.

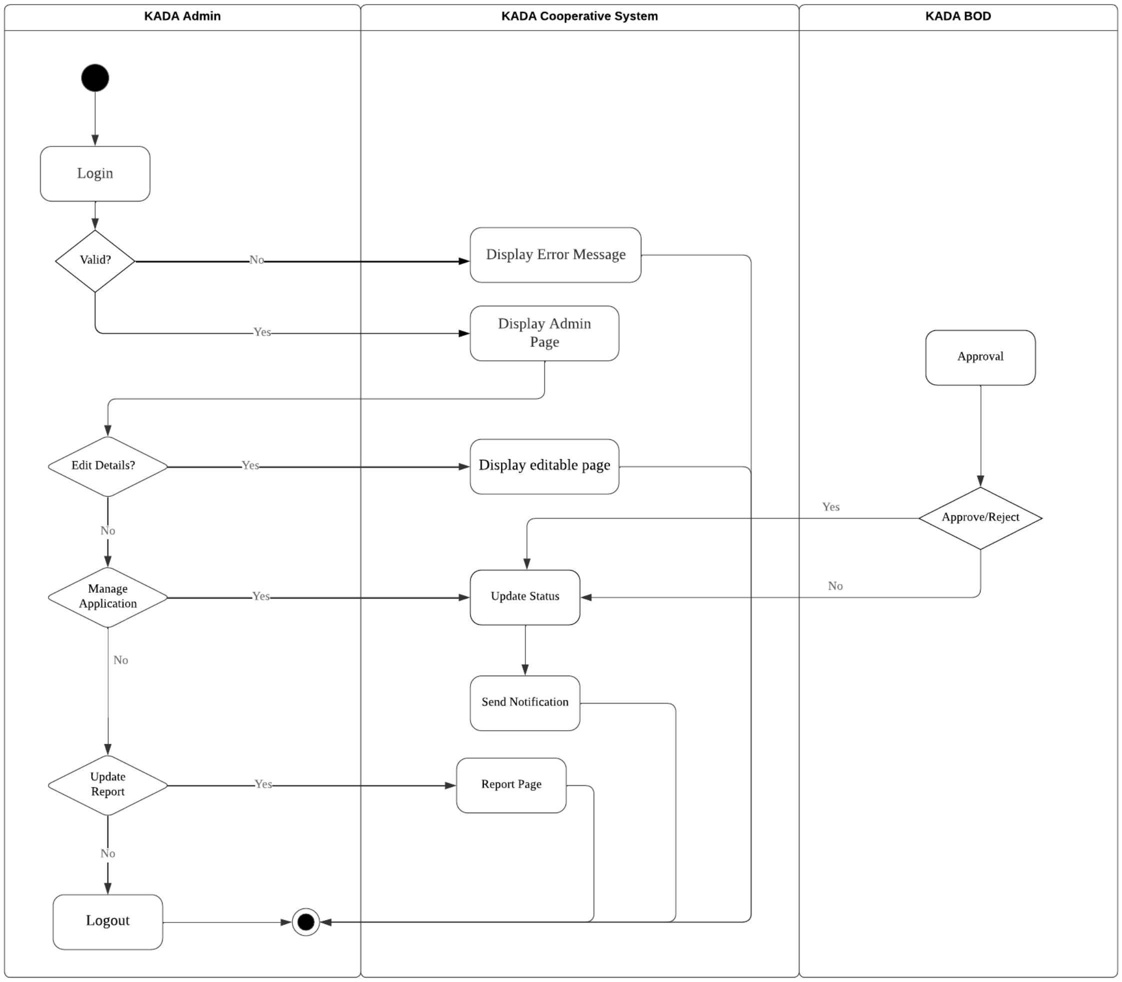


Figure 2.3.6.1: Activity Diagram for Manage Member

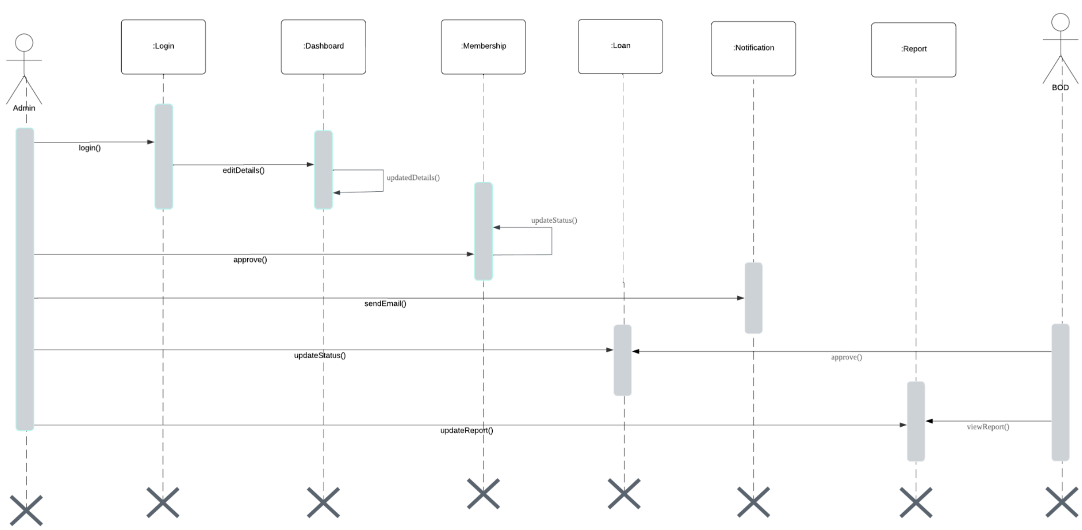


Figure 2.3.6.2: Sequence Diagram for Manage Member

2.3.9 UC009: Use Case <Review Application>

Table 2.3.9.1: Use Case Description for Review Applications

Use case: Review Applications
ID: UC009
Actors: KADA Cooperative Boards of directors KADA Cooperative Admin
Preconditions: 1. KADA cooperative BOD and admin are required to log into the system before reviewing applications.
Flow of events: 1. If BODs want to review member registration applications. 1.1. BODs select the “Permohonan Ahli” button. 1.2. The system will display lists of member applications. 1.3. BODs will approve/reject applications after offline discussion. 2. Else if BODs want to review loan applications. 2.1. BODs select the “Permohonan Pembiayaan” button. 2.2. The system will display lists of loan applications. 2.3. BODs will approve/reject applications after offline discussion. 3. KADA cooperative admin will monitor the reviewing process to ensure all decisions are successfully submitted into the system.
Postconditions: 1. After the application has been reviewed, the system will send an email to notify KADA staff. 2. Admin can send email manually if automatic email is not sent to the applicant.
Alternative flow 1: 1. If BODs select an application, the system will display the application details.
Postconditions: 1. After the application has been reviewed, the system will send an email to notify KADA staff. 2. Admin can send email manually if automatic email is not sent to the applicant.

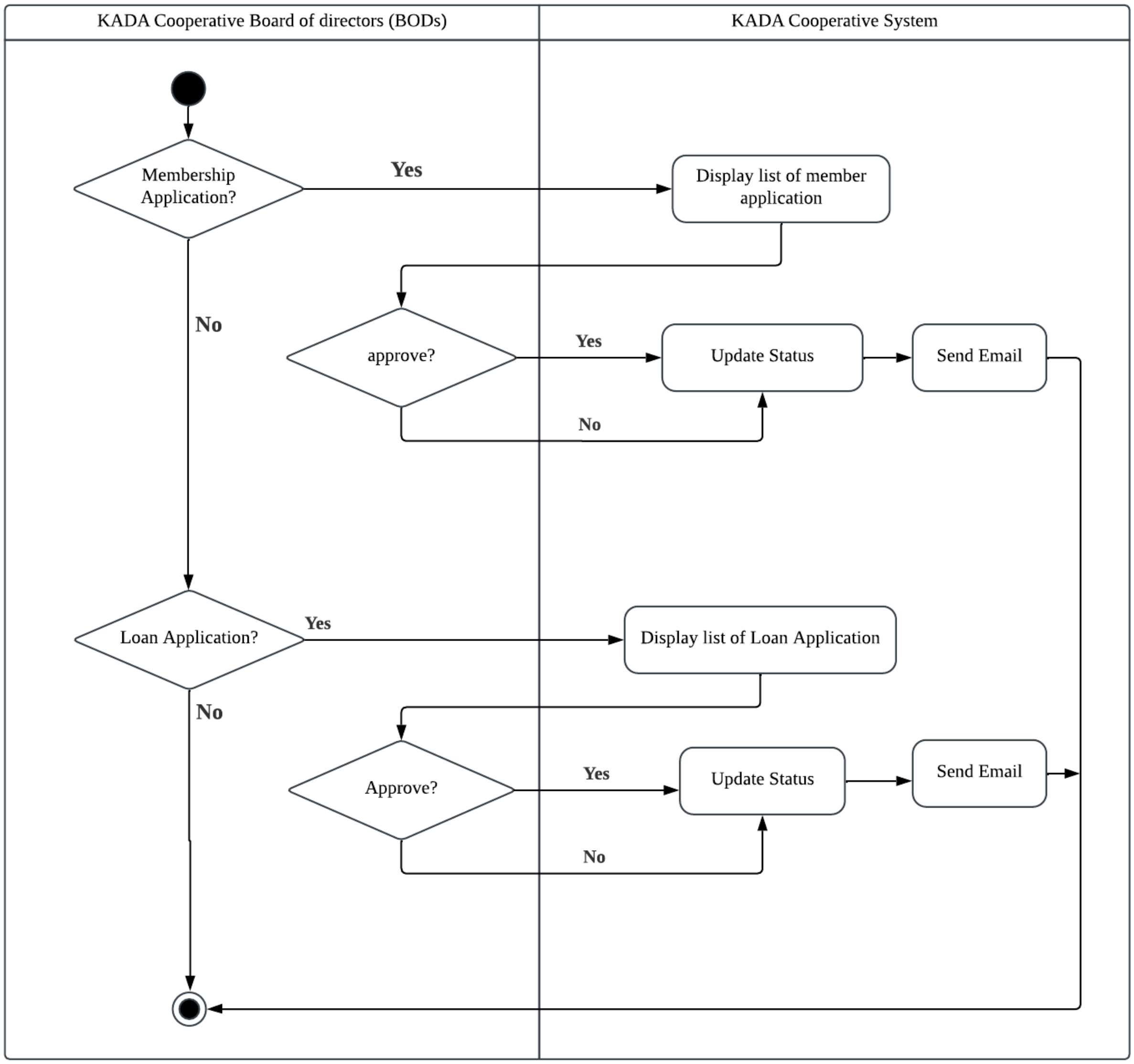


Figure 2.3.9.1: Activity Diagram for Review Applications

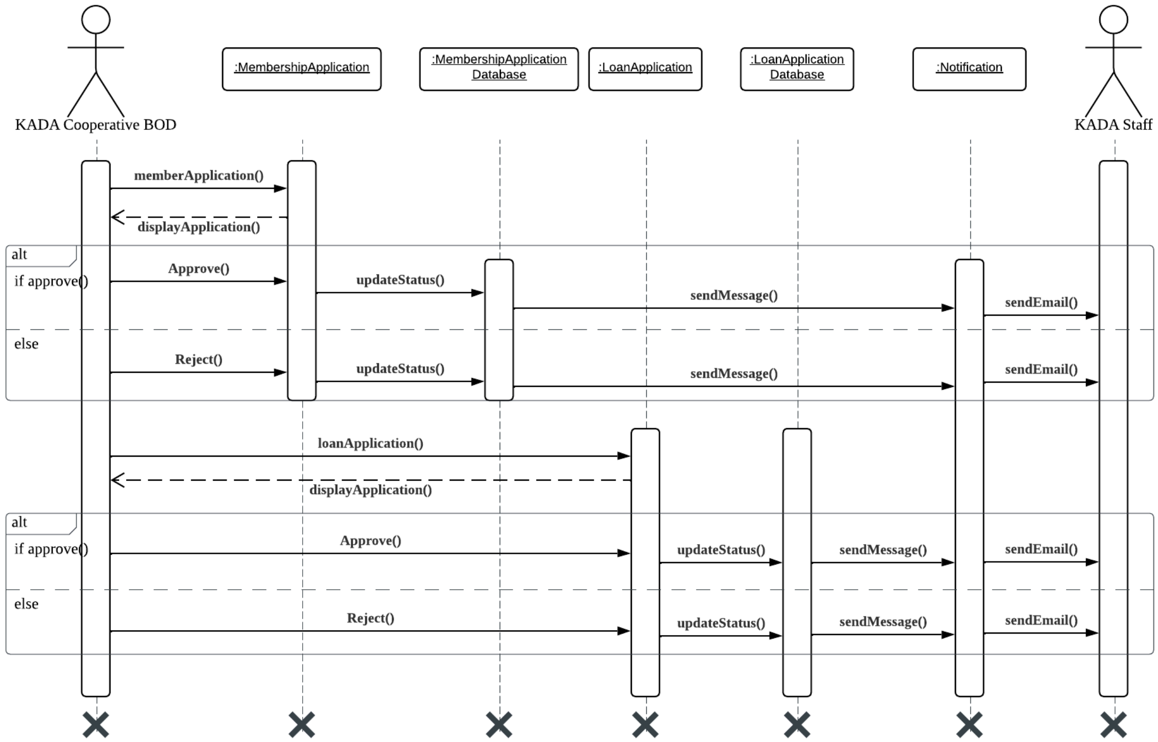


Figure 2.3.9.2: Sequence Diagram for Review Applications

2.4 **Performance and Other Requirements**

This section details the non-functional requirements, which have been categorized into Software System Attributes, Performance Requirements, and Other Requirements, adapted to the needs of the KADA Cooperative System.

1. Software System Attributes

- Usability:
  - ➢ The system shall provide a user-friendly interface for all user roles, enabling administrators, board members, and members to interact efficiently.
  - ➢ KADA Specific: Members can easily view personal information and submit loan applications, while administrators and board members can manage approvals and generate reports seamlessly.

- Reliability:
  - ➢ Ensure consistent performance of key features such as dashboards, loan applications, and reporting.
  - ➢ KADA Specific: Loan processing and real-time transactions must operate reliably, with minimal downtime, to maintain user trust and system efficiency.

- Maintainability:
  - ➢ Policy parameters are interest rates, repayment terms, membership benefits, or others that need to be changed within 10 minutes without downtime.
  - ➢ KADA Specific: Provide the administrators with a settings interface to quickly update the policies, so that the system could adapt to the needs of the cooperative without interruption.

- Portability:
  - ➢ The system shall support desktops, tablets, and smartphones and provide a consistent responsive interface across devices.
  - ➢ KADA Specific: Mobile access for board members ensures remote loan application approvals, while employees access via desktop during business hours.

- Compatibility:

- ➢ The system should integrate well with payroll systems for efficient loan repayments.
- ➢ KADA Specific: Payroll deductions for loan repayments should work seamlessly with the organization's existing systems, minimizing manual interventions.

2. Performance Requirements

- Response Time:
- ➢ Authentication and loading of the dashboard should take less than 2 seconds, while member data or loan application retrieval should take less than 3 seconds.
- ➢ KADA Specific: Ensures smooth interaction for all users, particularly during busy cooperative operations.

- Throughput:
- ➢ The system shall be able to process over 50 transactions per second at peak periods.
- ➢ KADA Specific: Allows real-time recording of loan applications and approvals during high demand.

- Capacity:
- ➢ The system will support 1,000 concurrent users and store data for 100,000 members without degradation of system performance.
- ➢ KADA Specific: Supports current volume of users and scales to grow with membership.

- Availability:
- ➢ System shall be available for use 99.9%, permitting only 8 hours of downtime annually.
- ➢ KADA Specific: Ensures cooperative services will be up during work hours to meet the needs of members and the organization.

3. Other Requirements

- Security:
- ➢ The system shall ensure data security by employing AES-256 encryption, multi-factor authentication for sensitive operations.
- ➢ KADA Specific: Protects member data, loan applications, and board-level reports from unauthorized access or breaches.

- Safety:
  - ➢ Unauthorized access shall not be allowed via RBAC, audit trails tracking the activities for accountability.
  - ➢ KADA Specific: Only the admin and board members will have access to sensitive data for member trust and operational security.

- Legal and Regulatory:
  - ➢ Compliance with data protection legislation, such as PDPA in Malaysia, and the ISO/IEC 27001 standard, must be ensured.
  - ➢ KADA Specific: Ensures that all cooperative data and transactions are legally complied with to avoid penalties and ensure member privacy.

- Environmental:
  - ➢ Optimize CPU and memory usage; ensure they do not exceed 75% during normal operations.
  - ➢ KADA Specific: Ensures system sustainability by avoiding overuse of computational resources to support long-term cooperative goals.

2.5 **Design Constraints**

a) *Environmental constraints*
- The system is designed to be able to operate under the specific temperature and humidity levels which are 0^0C to 40^0C and 20% to 80%.
- The system is able to function well in any network conditions which include bandwidth limitation situations to ensure the system remains accessible by the users.

b) *Hardware constraints*
The minimum hardware specifications that are needed to be able to run the system are describing as following:
- Memory: a minimum of 8 GB of RAM
- Processor: 2.4 GHz multi-core
- Storage: at least 500 GB of SSD storage

c) *Security Constraints*
- The system implements a Role-Based Access Control (RBAC) to restrict users access to the sensitive data based on their roles within the cooperative system.
- Sensitive members and financial data must be stored and protected through data encryption using AES-256 and secure access protocols.

d) *Compatibility constraints*
- The system can function well across multiple operating systems such as Microsoft Windows, macOS, and Linux and various types of devices without need to modify the system.

e) *Performance constraints*
- Each user has a maximum response time of 3 seconds under normal load conditions which is crucial to maintain the user satisfaction of the system.
- The system has to support up to 500 users to be able to access and interact with the system simultaneously without any performance degradation.
- The response time for major transactions such as the loan process is not more than 2 seconds.

3. Architectural Rationale

This section of the SD describes the architectural style and the rationale or justification of the system.

3.1 Architecture Style and Rationale

Architectural Rationale

The architecture of the KADA Online System is based on the Laravel Framework, a contemporary PHP-based web development framework that uses the Model-View-Controller design pattern. This provides a structured, maintainable, and scalable system architecture.
Architecture Style and Rationale

- Authentication and Authorization
  - ➢ Laravel simplifies user authentication and authorization. Pre-built authentication scaffolding allows the developer to easily implement login, registration, and password reset functionality. RBAC can be efficiently managed using middleware.

- Support for MVC out of the Box
  - ➢ Laravel natively implements the MVC architecture, which promotes a clean separation of concerns between the Model (data), View (user interface), and Controller (business logic). This separation enhances code maintainability, supports modular development, and allows for better scalability as the system grows.

- Security and Encryption
  - ➢ Laravel puts much emphasis on security, with features such as CSRF protection, and SQL injection prevention. The framework also supports secure cookies and encrypted session data, ensuring data privacy and protection.

- Blade Templating Engine
  - ➢ The Blade templating engine allows the creation of dynamic and reusable views with simple syntax. It supports control structures, such as loops and conditionals, and template inheritance, which simplifies the development of user interfaces.

- Cross-Browser Compatibility
  - ➢ The Blade Templating Engine creates responsive cross-browser compatible front-end components. That means that the system's user interface works in major web browsers, such as Google Chrome, Mozilla Firefox, and Microsoft Edge, offering consistent users' experience.

- Scalability and Extensibility
  - ➢ The modular architecture that Laravel provides accommodates scaling with ease. The system that will be built for KADA has the potential to optimize even more when handling bigger-sized data and more users that access the system concurrently. By using this framework, accommodating new features or modules wouldn't be that hard anymore.

- Clean Routing and Middleware

- ➢ The Laravel routing system generates clean and human-readable URLs for different modules of a system, like registration, activities, and reports. Middleware does the job for handling Role-Based Access Control (RBAC) and request-level security, ensuring only authorized access to certain system resources.

- Community Support and Rich Ecosystem
- ➢ The documentation is extensive, tutorials are in plenty, and there is great community-driven support within the active developer community in Laravel. Such an ecosystem gives way to rapid troubleshooting, access to best practices that speeds up development, hence more reliable systems.

Conclusion:
The reason for choosing Laravel as the base framework for the KADA Online System is its support for the MVC pattern, fast development, strong security, database management, scalability, and a large community. This makes the system maintainable, secure, and future-proof for changes in the requirements of KADA RS and its users. give me other features about it

4. Architectural Views

This chapter describes the architectural views of the system based on the framework developed by Philippe Kruchten. The views consist of scenario view, logical view, process view, development view and physical (deployment) view as shown in Figure 4.1. Each view is described in detail in table 4.1.

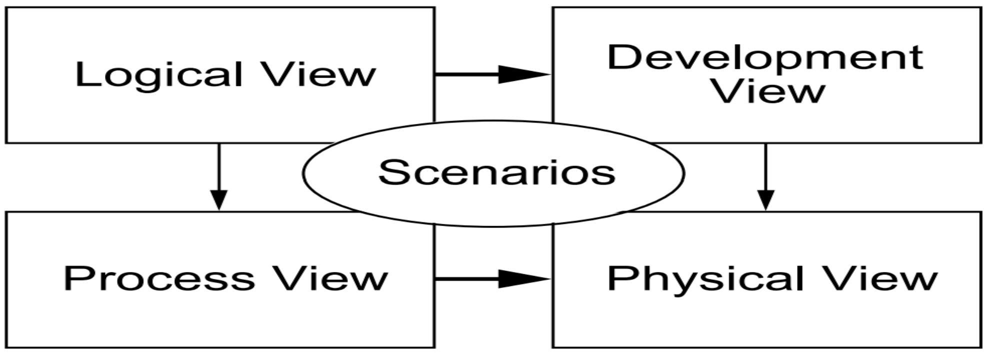


Figure 4.1: Architectural Views

Table 4.1: Views description and diagram used

Architectural Views	Description	Diagrams used
Use Case View	To describe the system's key users and how it interacts with system function and components.	Use Case Diagram
Logical View	To represent system's static structure	Class Diagram
Process	This is to represent how the object or components of the system interact with each other to perform specific functionality.	Sequence Diagram
Development (or Implementation) View	- To represent how the software components interact and are related to each other. - To group related classes, components, or other UML elements into logical packages.	Component Diagram Package Diagram
Physical (or Deployment) View	To show how the software component is working in a real-world environment.	Deployment Diagram

4.1 **Use Case View**

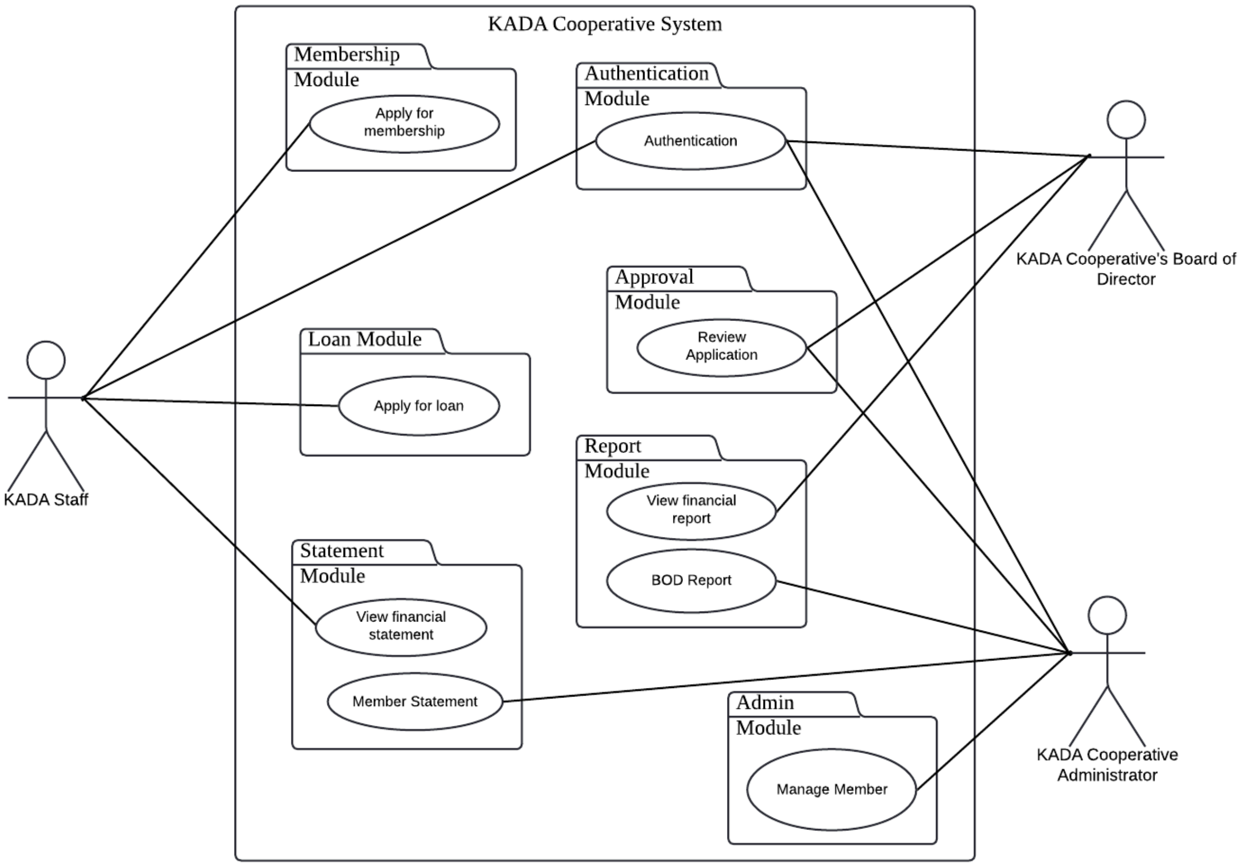


Figure 4.1.1: Use Case Diagram for <KADA Cooperative System>

The KADA Cooperative System will allow staff, administrators and the board of directors to manage membership, loans, approvals, financial reports and member statements through various interconnected modules, ensuring secure and efficient operations.

4.2 **Implementation View**

The implementation view for KADA Cooperative System focuses on the technical architecture and the process of converting the system's design into an executable software. It involves selecting the appropriate technologies, tools and frameworks that align with the system's requirements and constraints. This view is using the Model-View-Controller (MVC) architecture design, which outlines the integration of the system components to ensure they work well. The implementation view as shown in Figure 4.2.1 also considers the scalability and performance optimization to ensure the system can handle real-time efficiency.

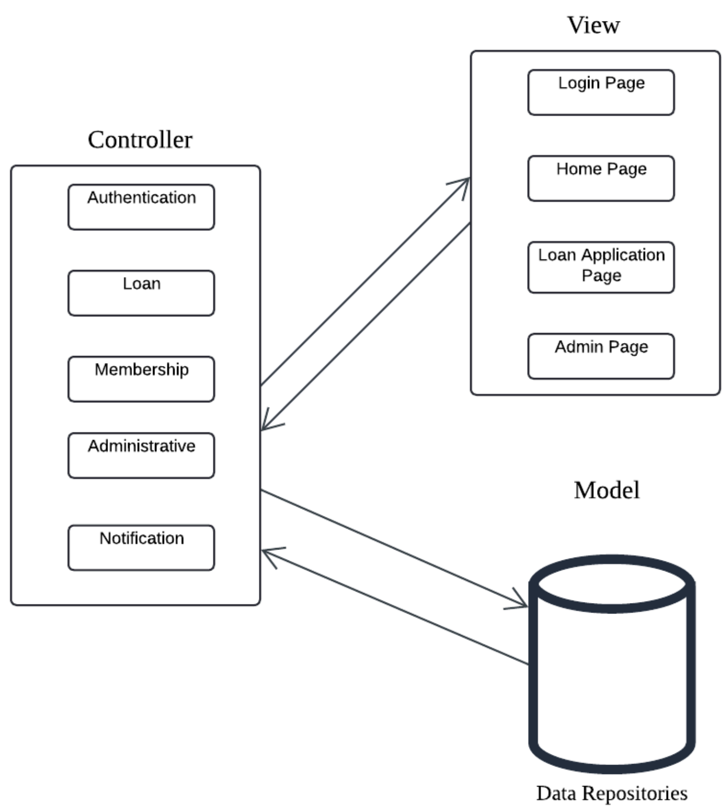


Figure 4.2.1: Architecture Diagram of KADA Cooperative System

The package diagram for the KADA Cooperative System visually organizes the system's components into related packages to show the structure between them. Each package represents a set of related modules that work together to fulfill the system functionality. The packet diagram is shown in Figure 4.2.2.

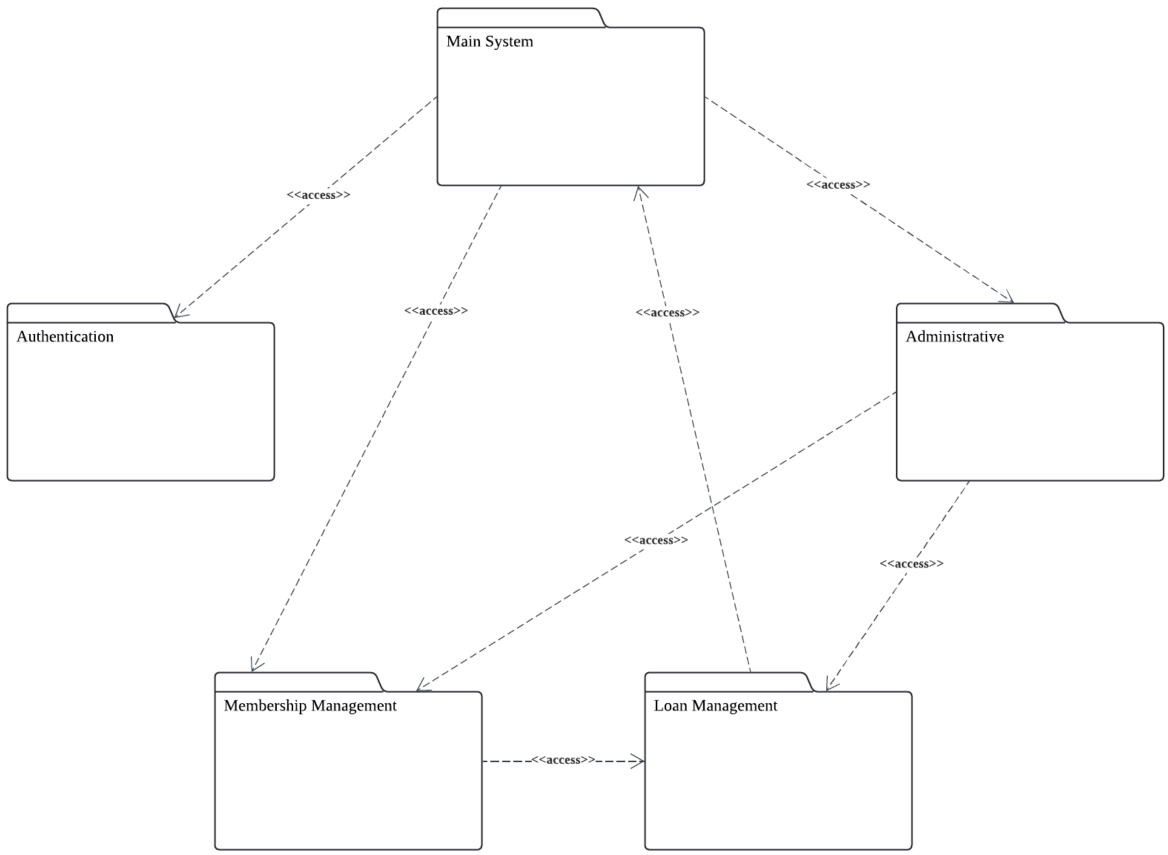


Figure 4.2.2: Package Diagram for KADA Cooperative System

Here also include the details of the packet diagram for each subsystem in the KADA Cooperative System.

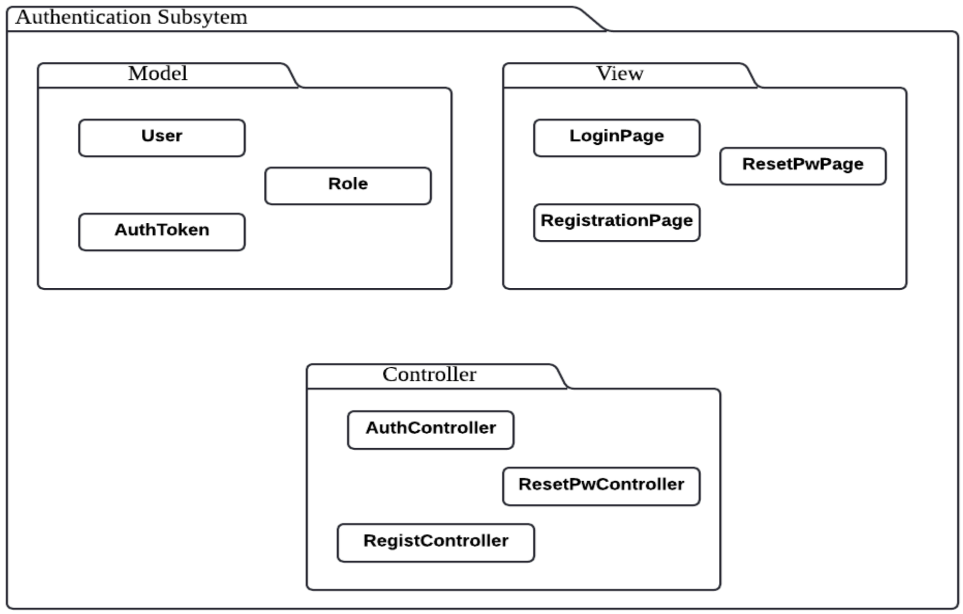


Figure 4.2.3: Package Diagram for <Authentication> Subsystem

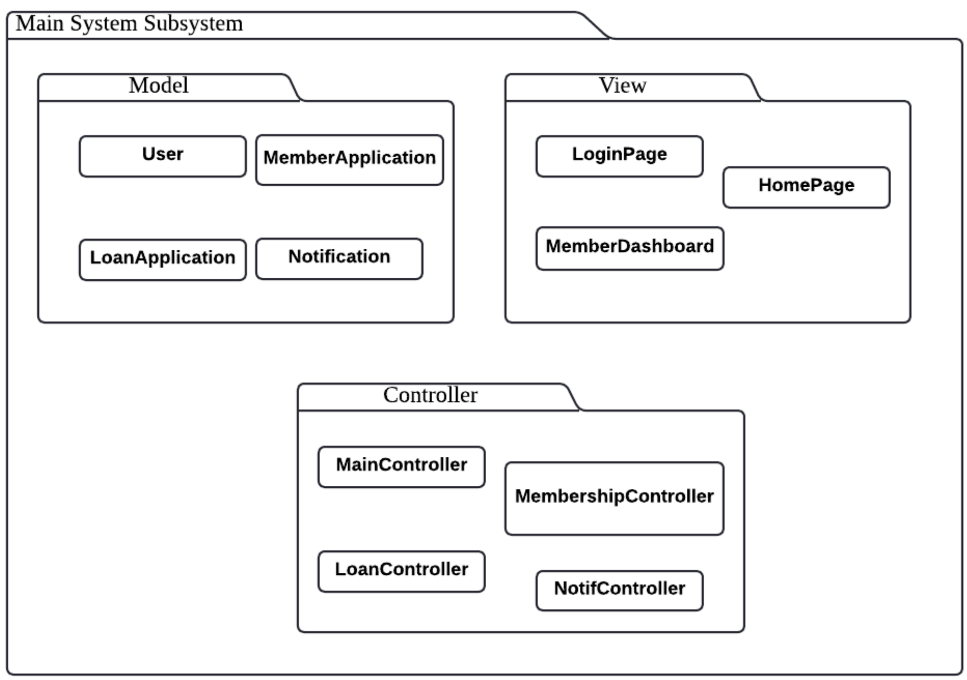


Figure 4.2.4: Package Diagram for <Main System> Subsystem

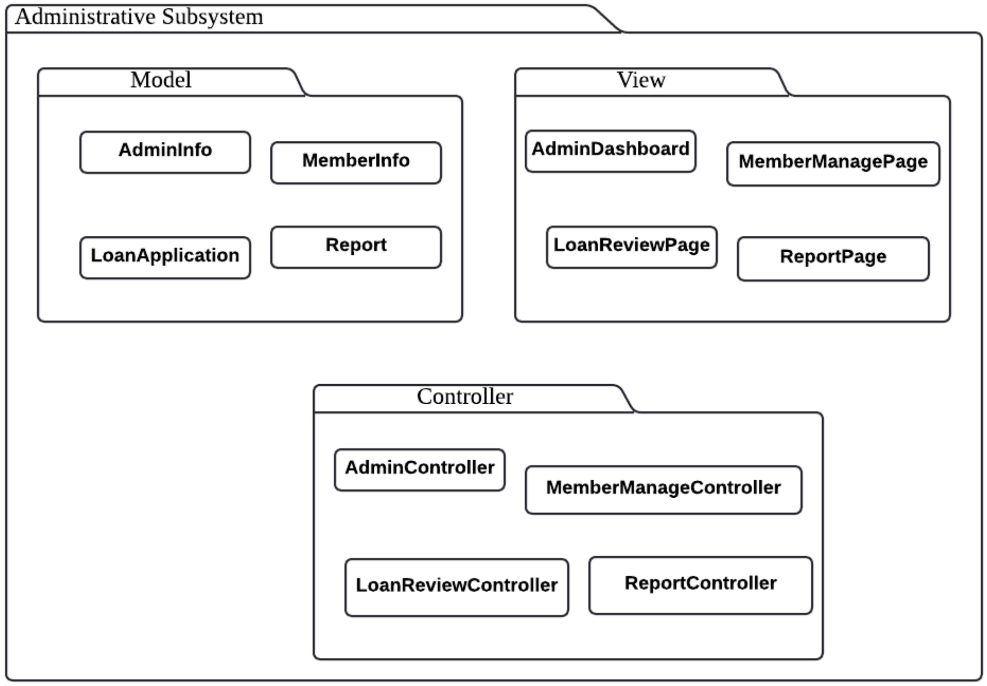


Figure 4.2.5: Package Diagram for <Administrative> Subsystem

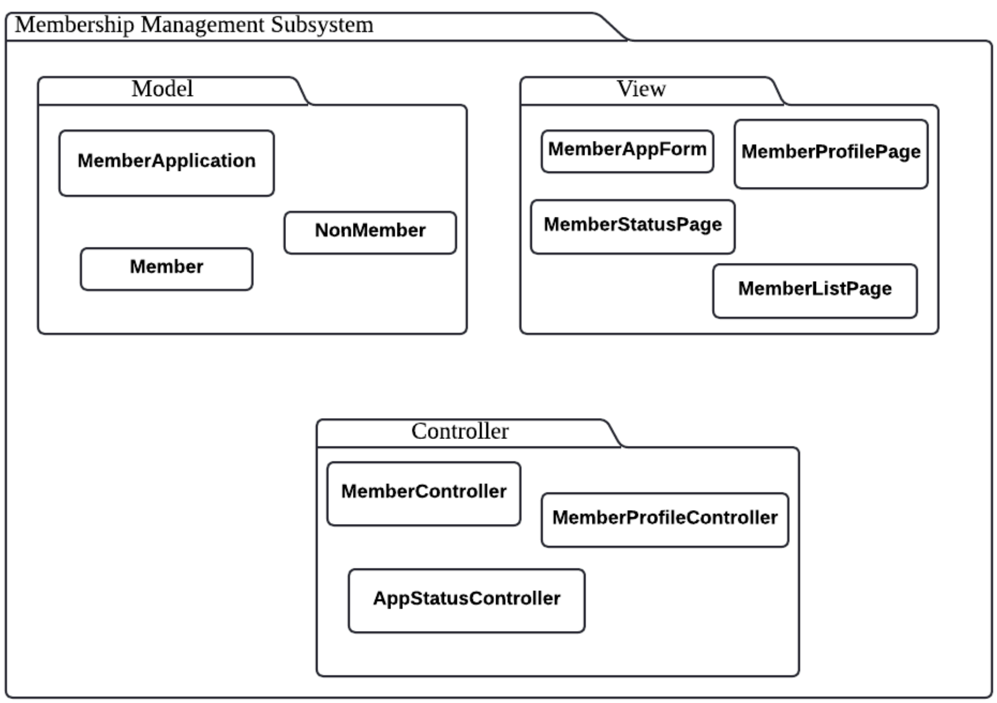


Figure 4.2.6: Package Diagram for <Membership Management> Subsystem

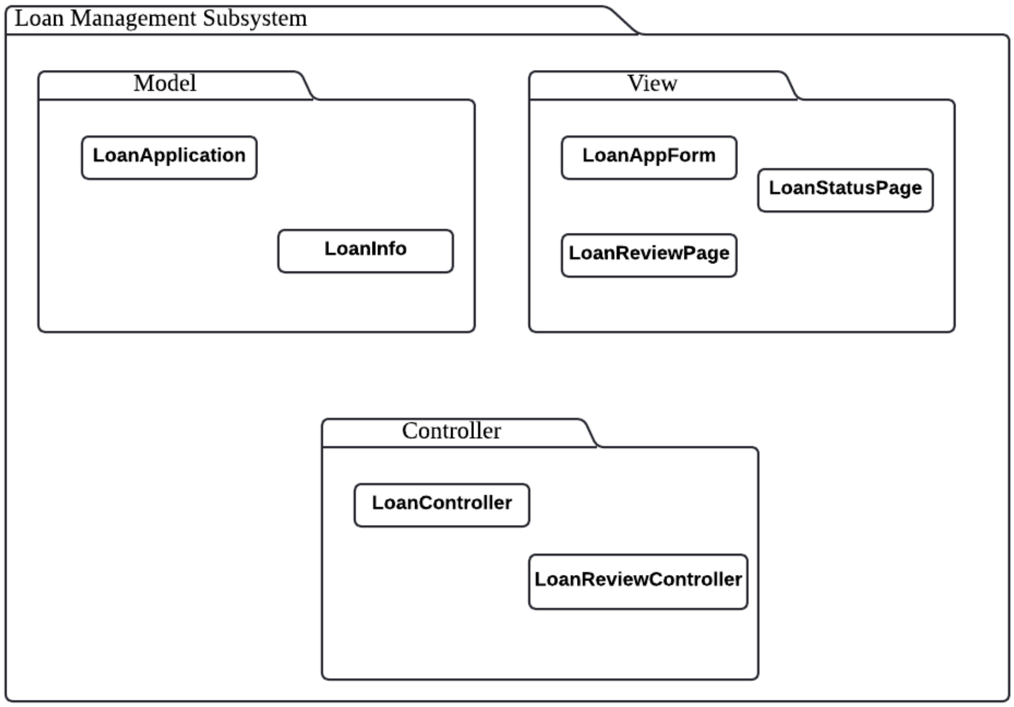


Figure 4.2.7: Package Diagram for <Loan Management> Subsystem

4.3 Logical View

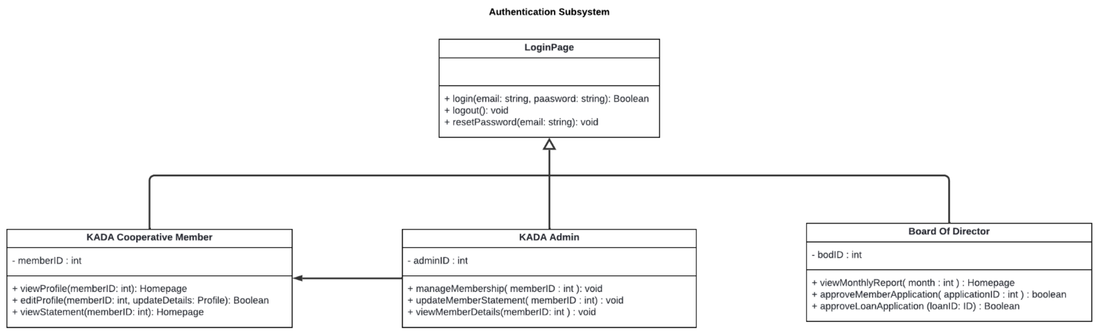


Figure 4.7: Class diagram for <Authentication> Subsystem

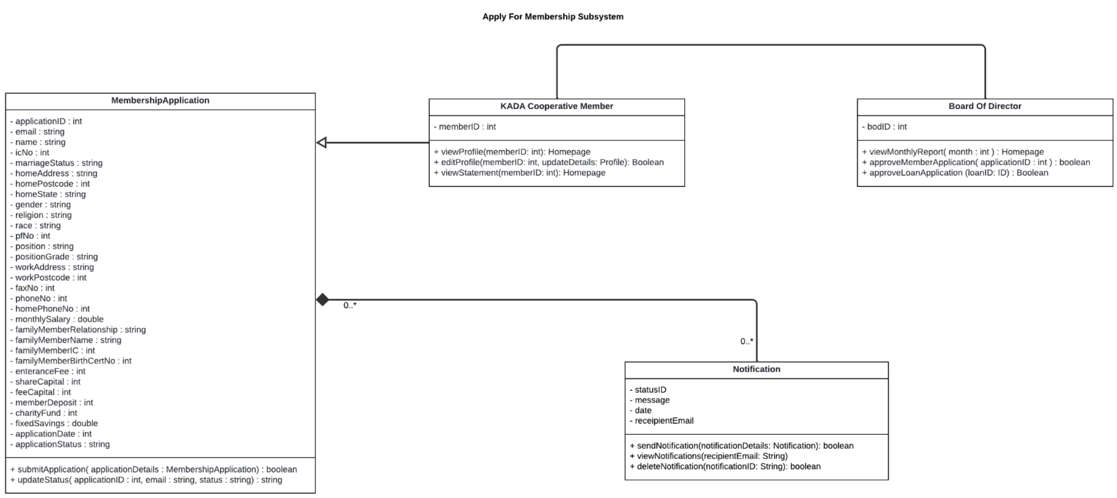


Figure 4.8: Class Diagram for <Apply For Membership> Subsystem

Apply For Loan Subsystem

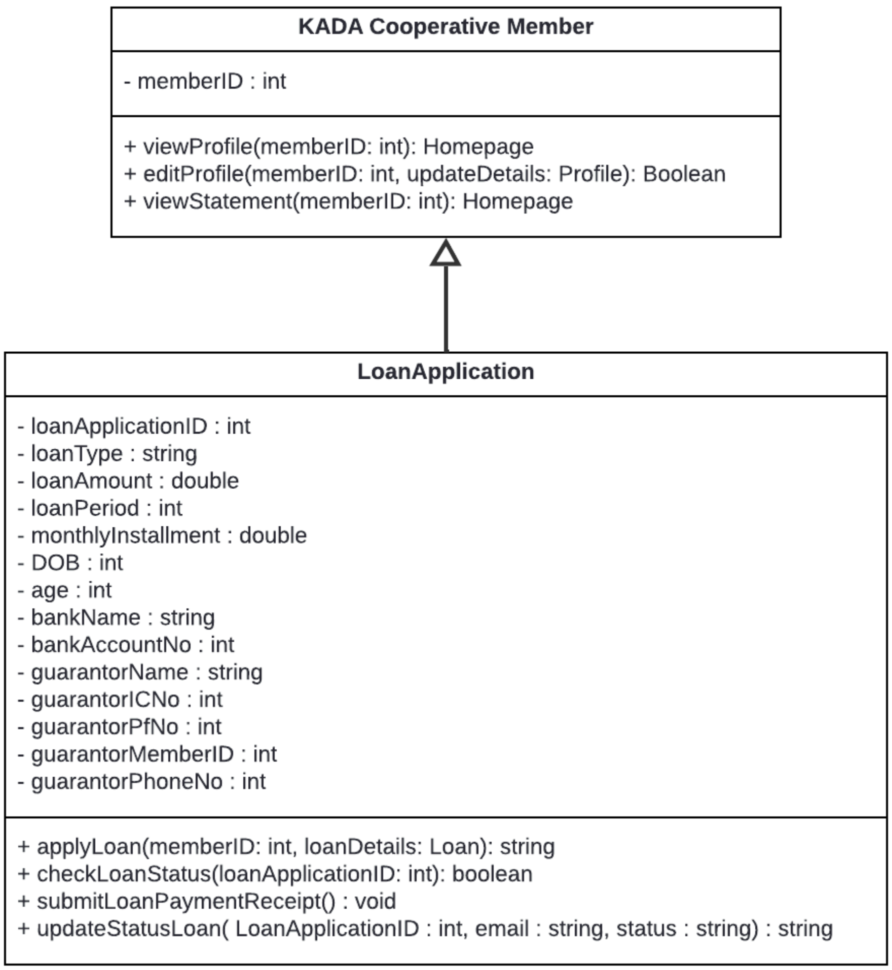


Figure 4.9: Class Diagram for <Apply For Loan> Subsystem

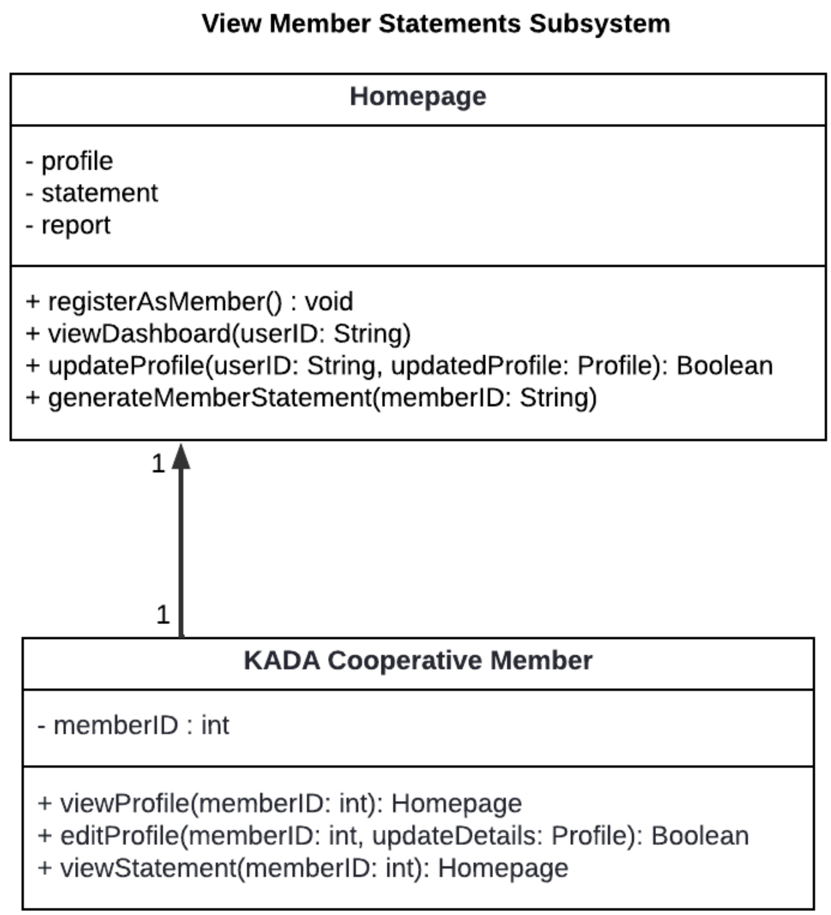


Figure 4.10: Class Diagram for <View Member Statement> Subsystem

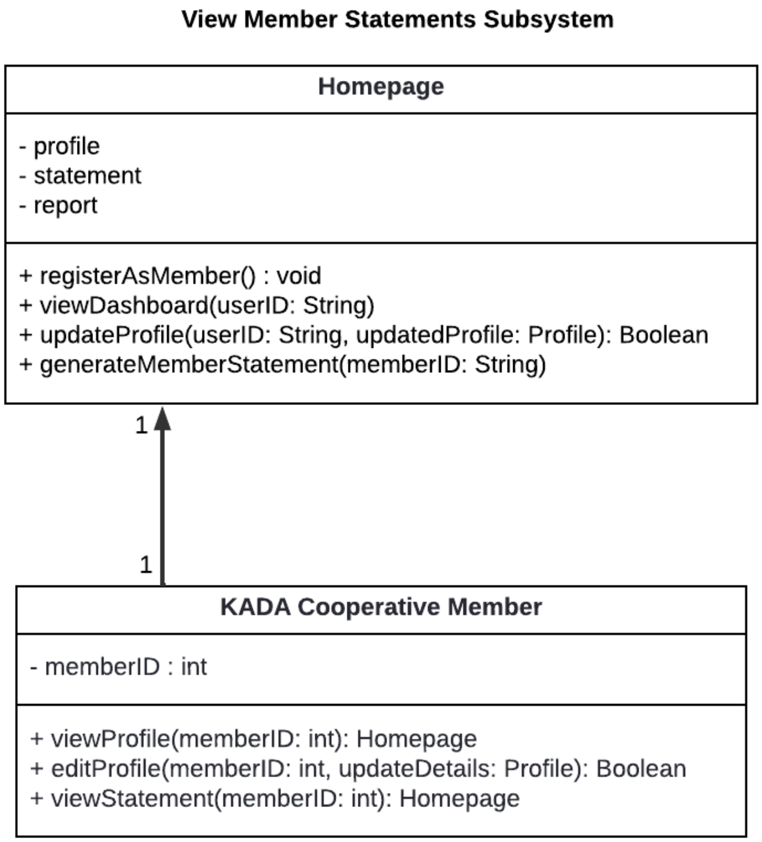


Figure 4.11: Class Diagram for <View Financial Statement> Subsystem

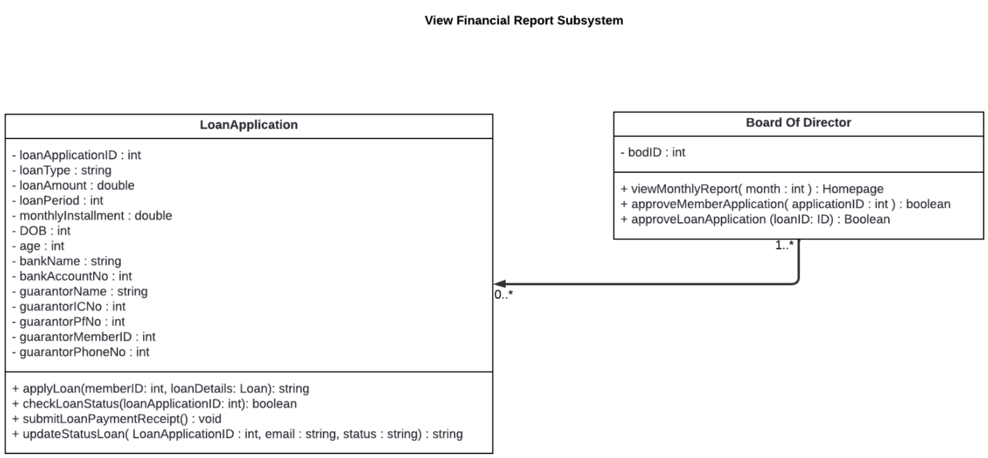


Figure 4.12: Class Diagram for <View Financial Report> Subsystem

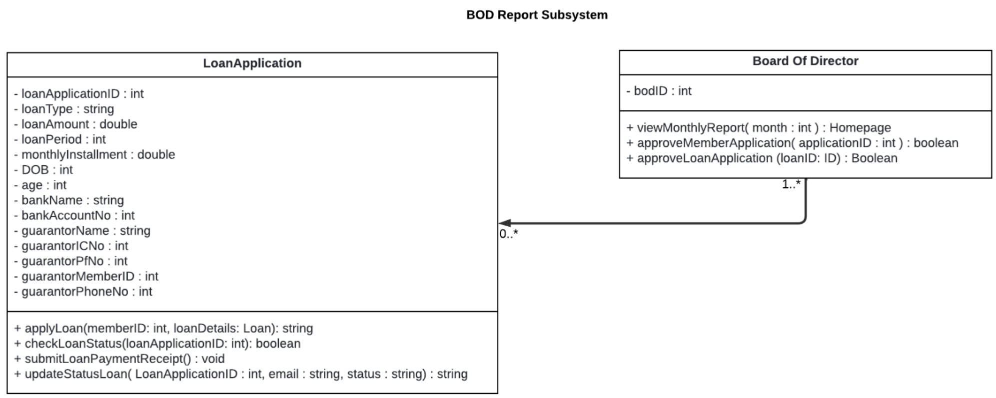


Figure 4.13: Class Diagram for <BOD Report> Subsystem

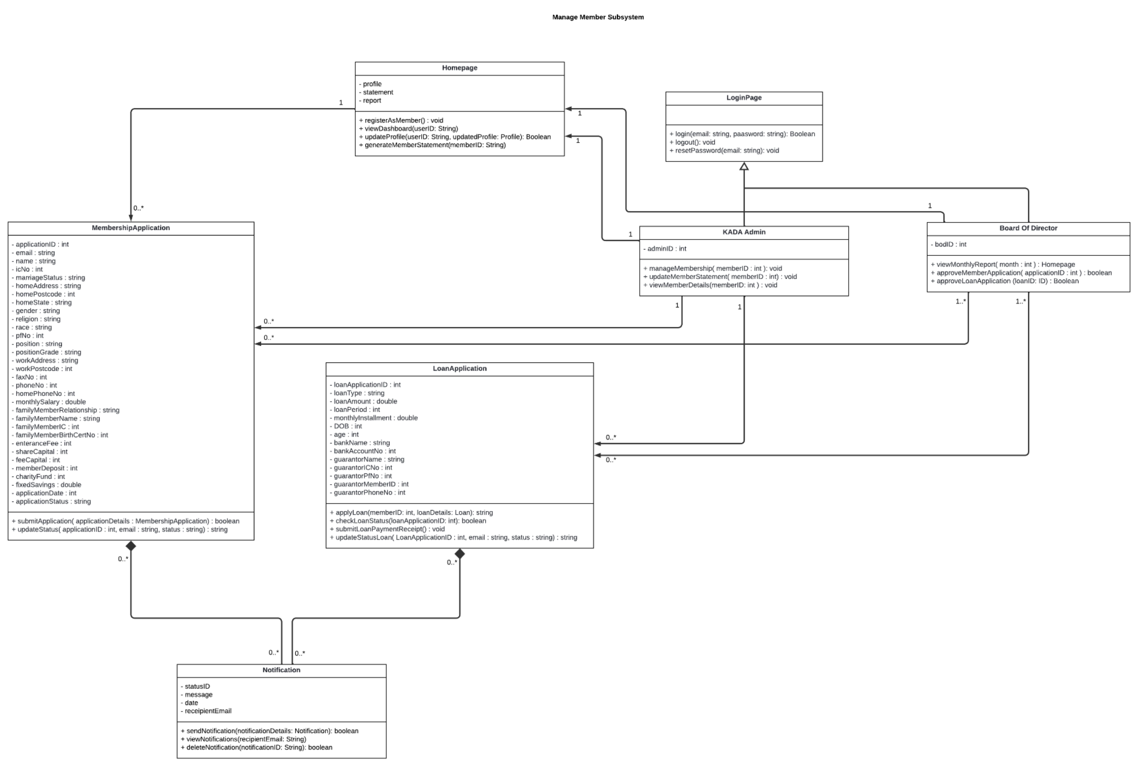


Figure 4.14: Class Diagram for <Manage Member> Subsystem

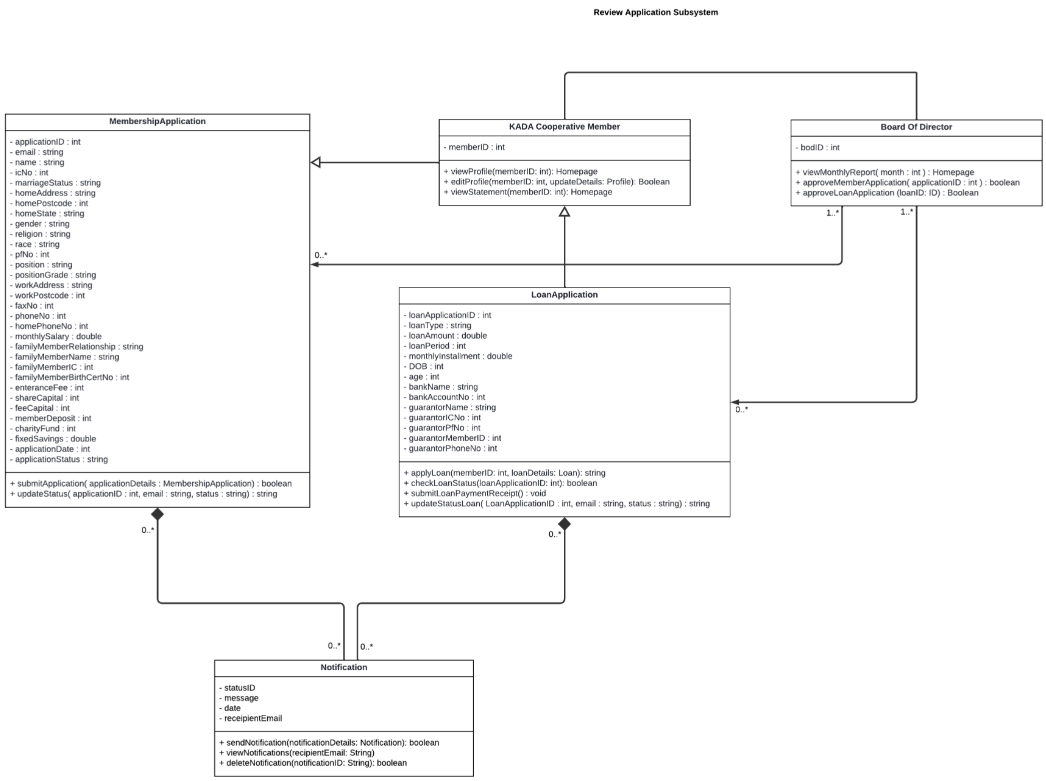


Figure 4.15: Class Diagram for <Review Application> Subsystem

4.4 Process View

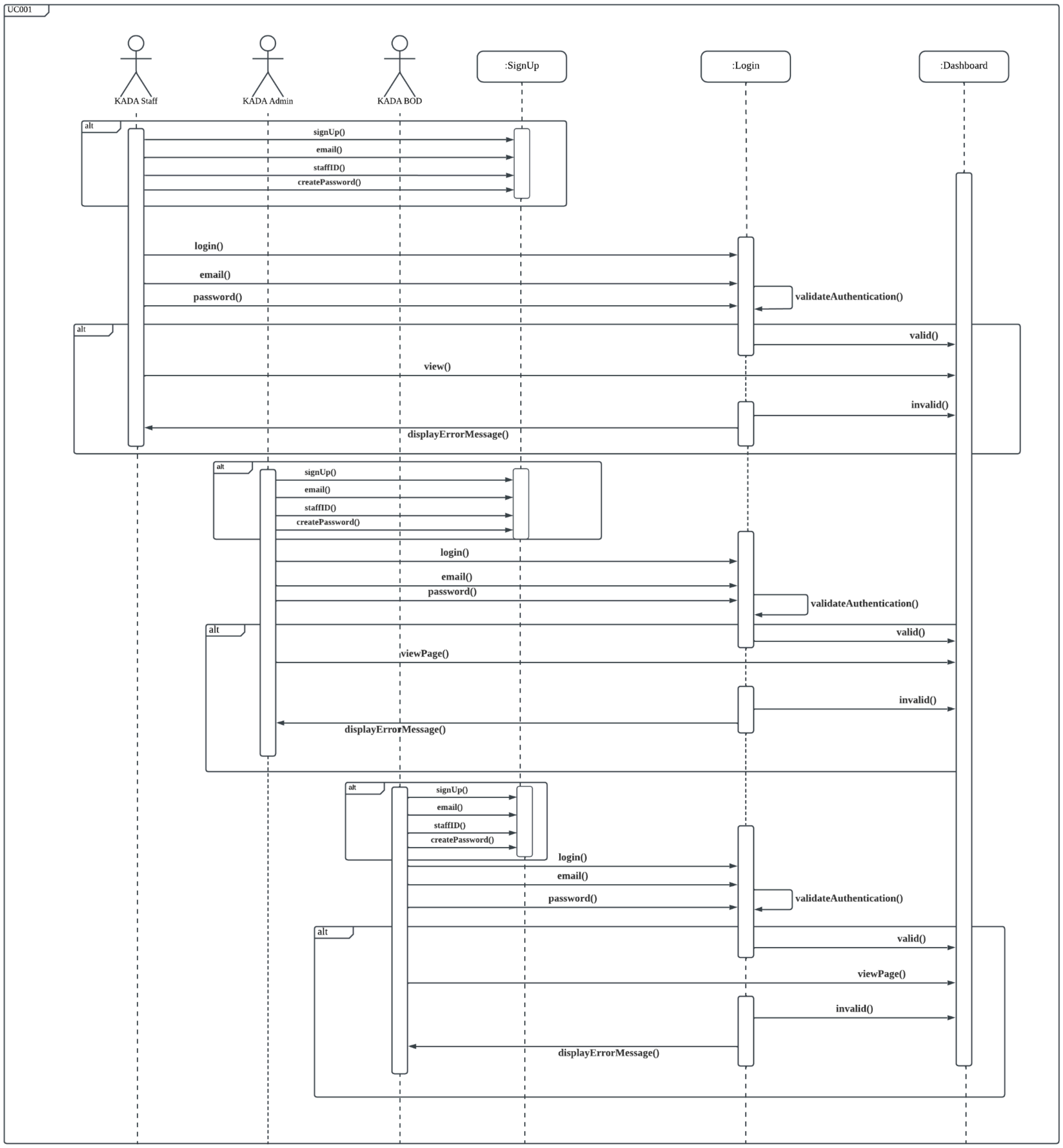


Figure 4.4.1: SD001 Sequence Diagram for <Authentication>

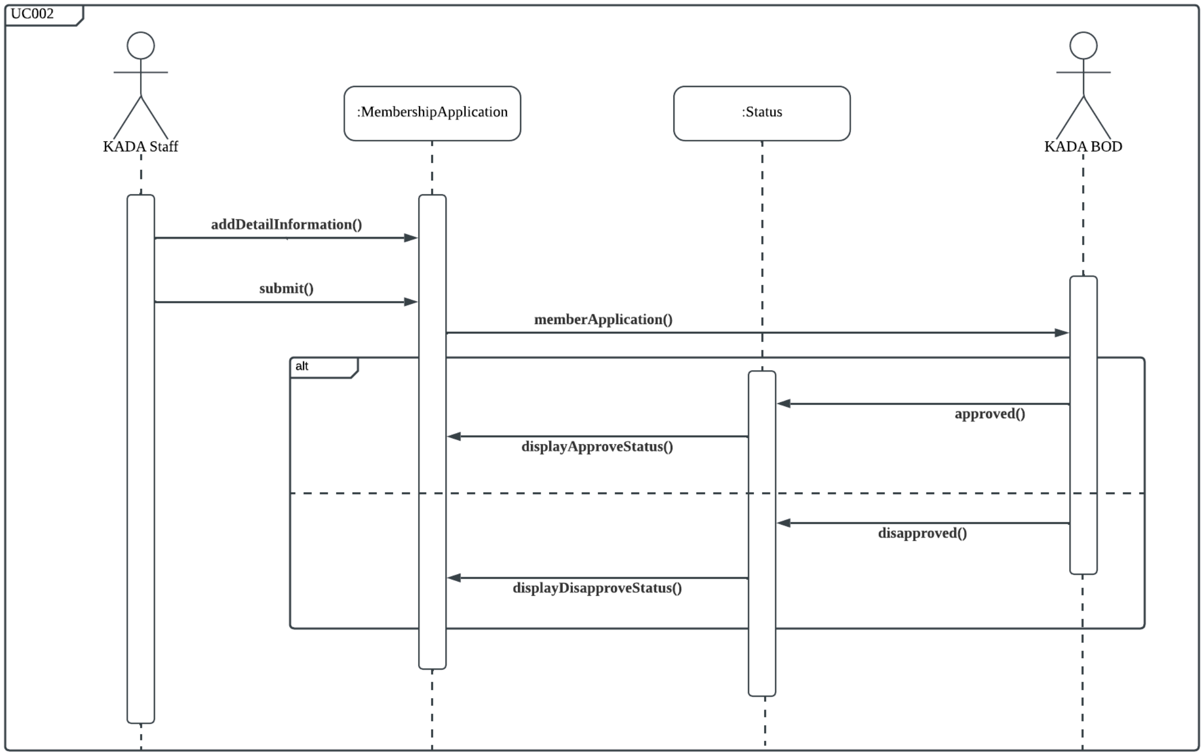


Figure 4.4.2: SD002 Sequence Diagram for <Apply for Membership>

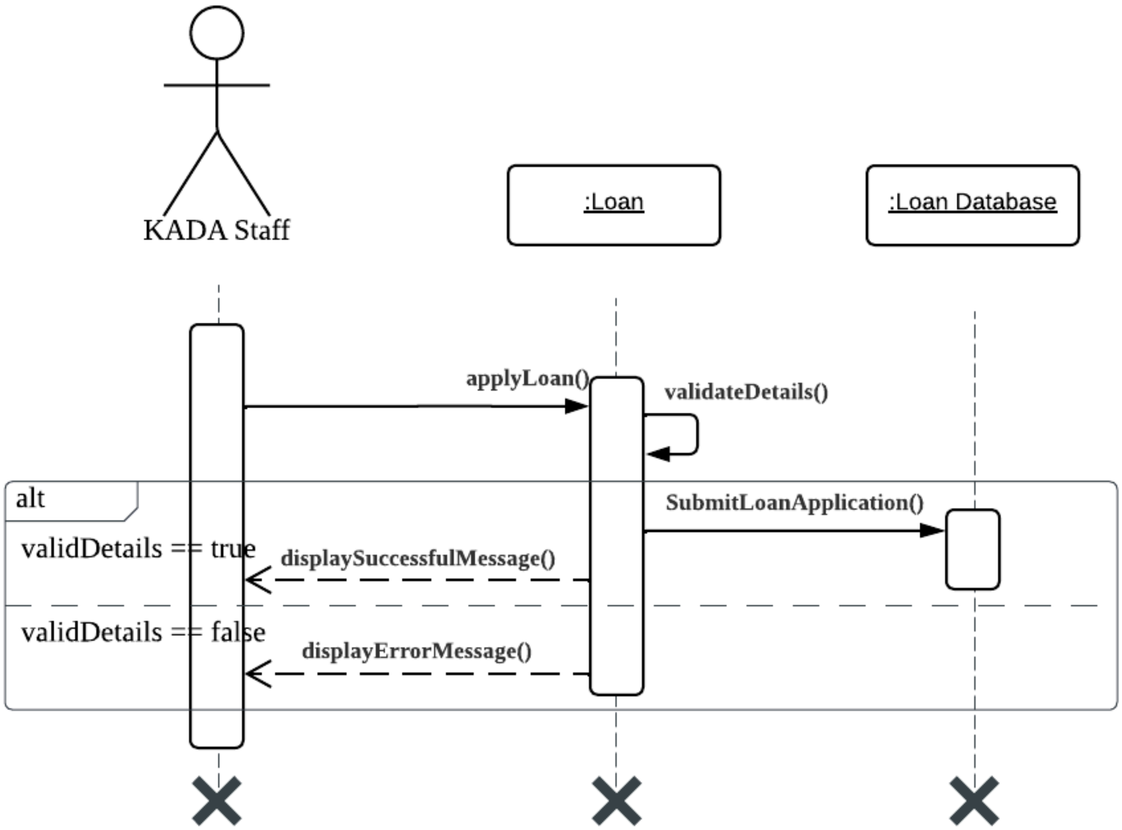


Figure 4.4.3: SD003 Sequence Diagram for <Apply for Loan>

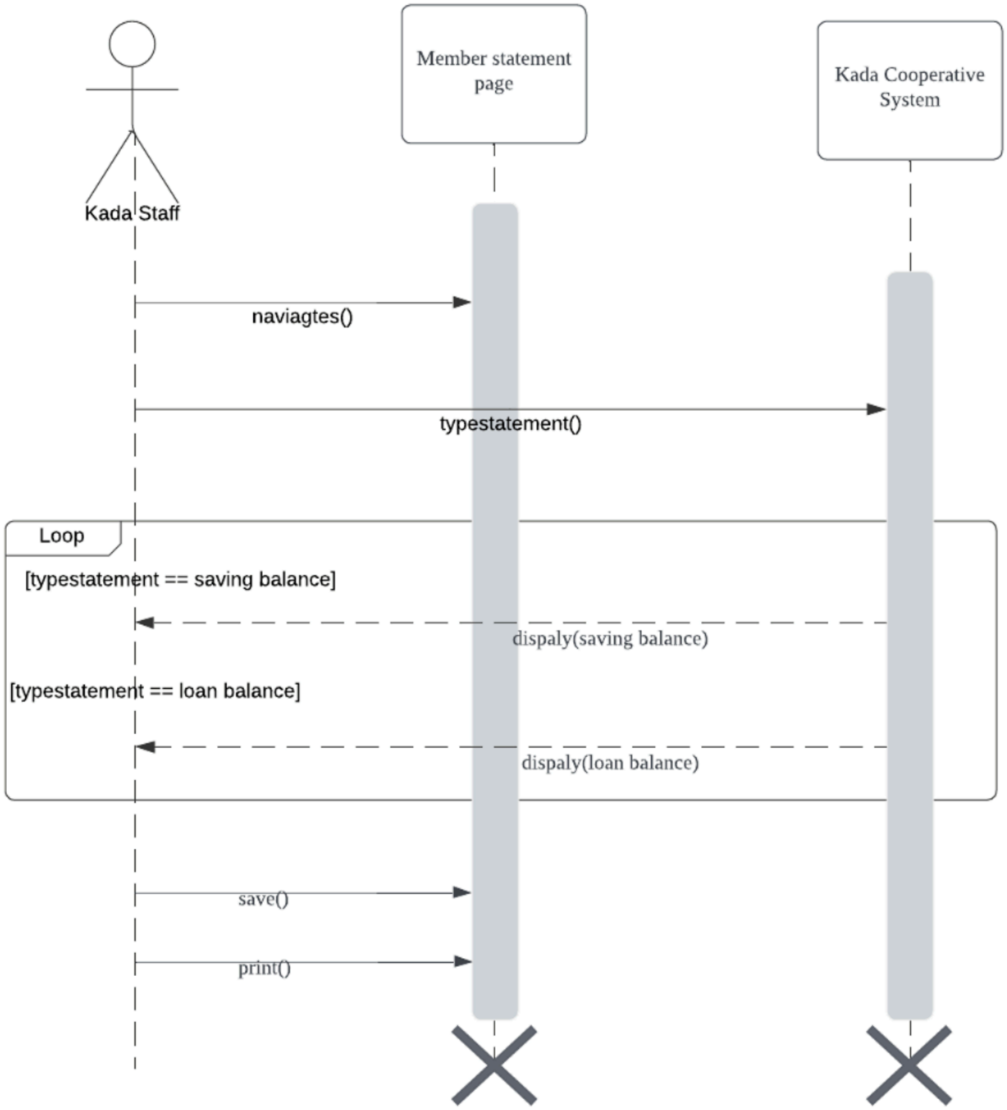


Figure 4.4.4: SD004 Sequence Diagram for <View Financial Statements>

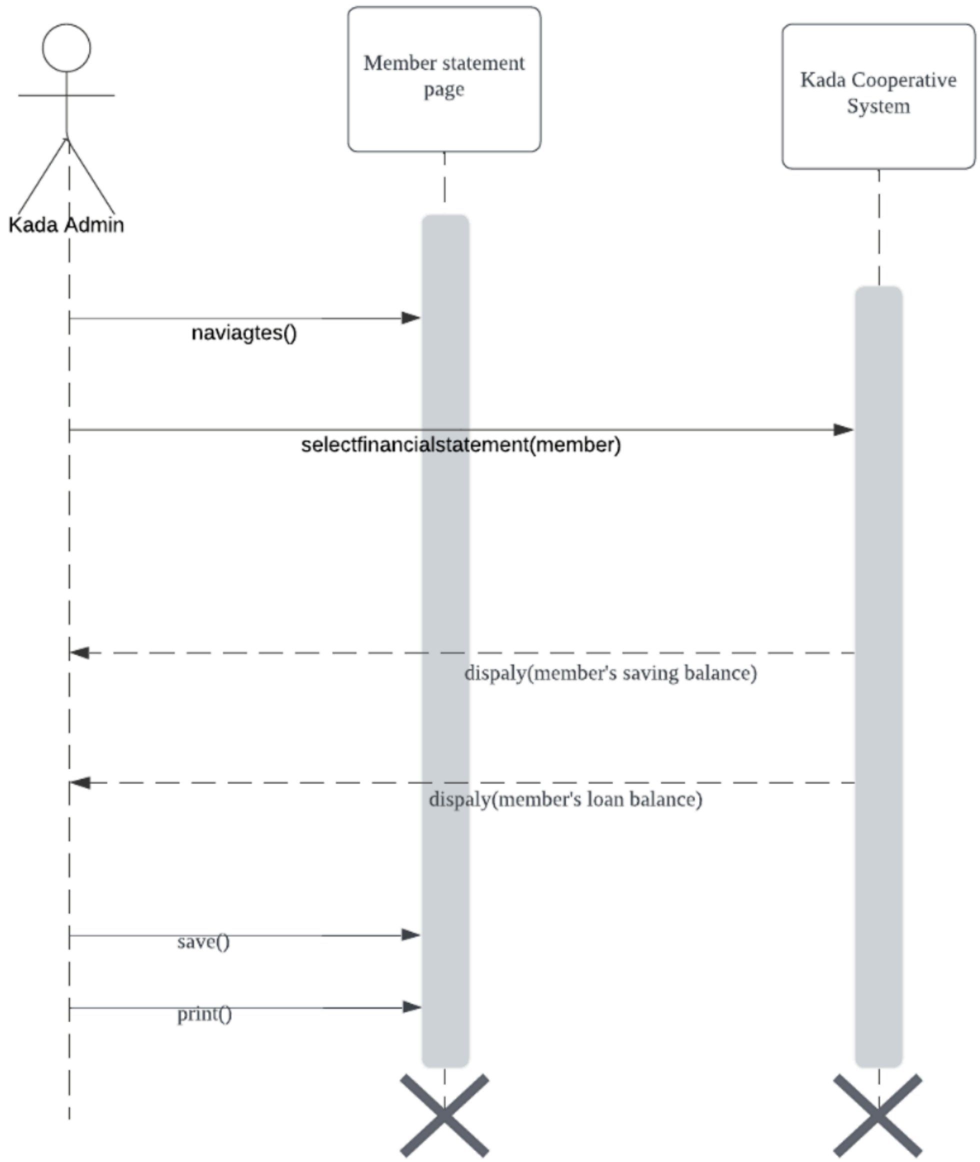


Figure 4.4.5: SD005 Sequence Diagram for <Member Statement>

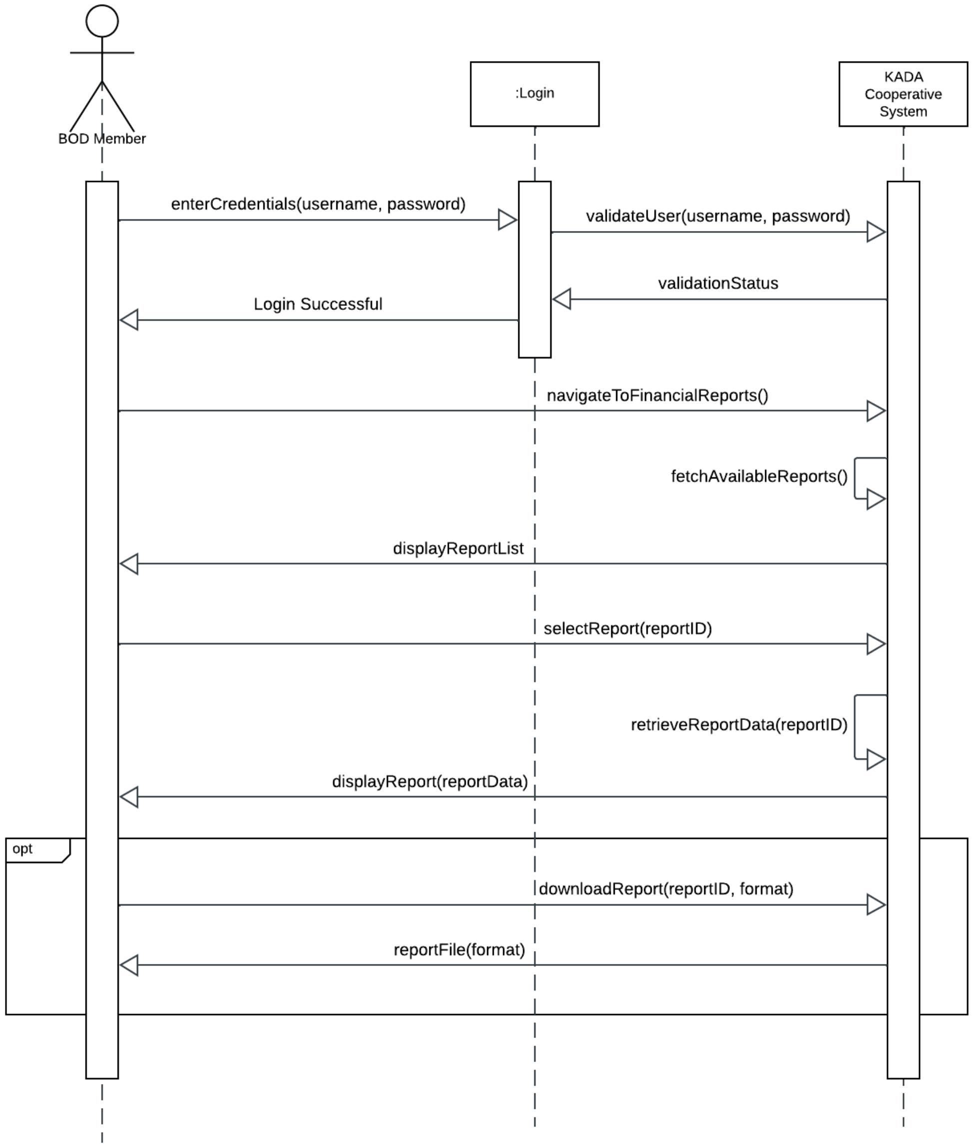


Figure 4.4.6: SD006 Sequence Diagram for <View Financial Reports>

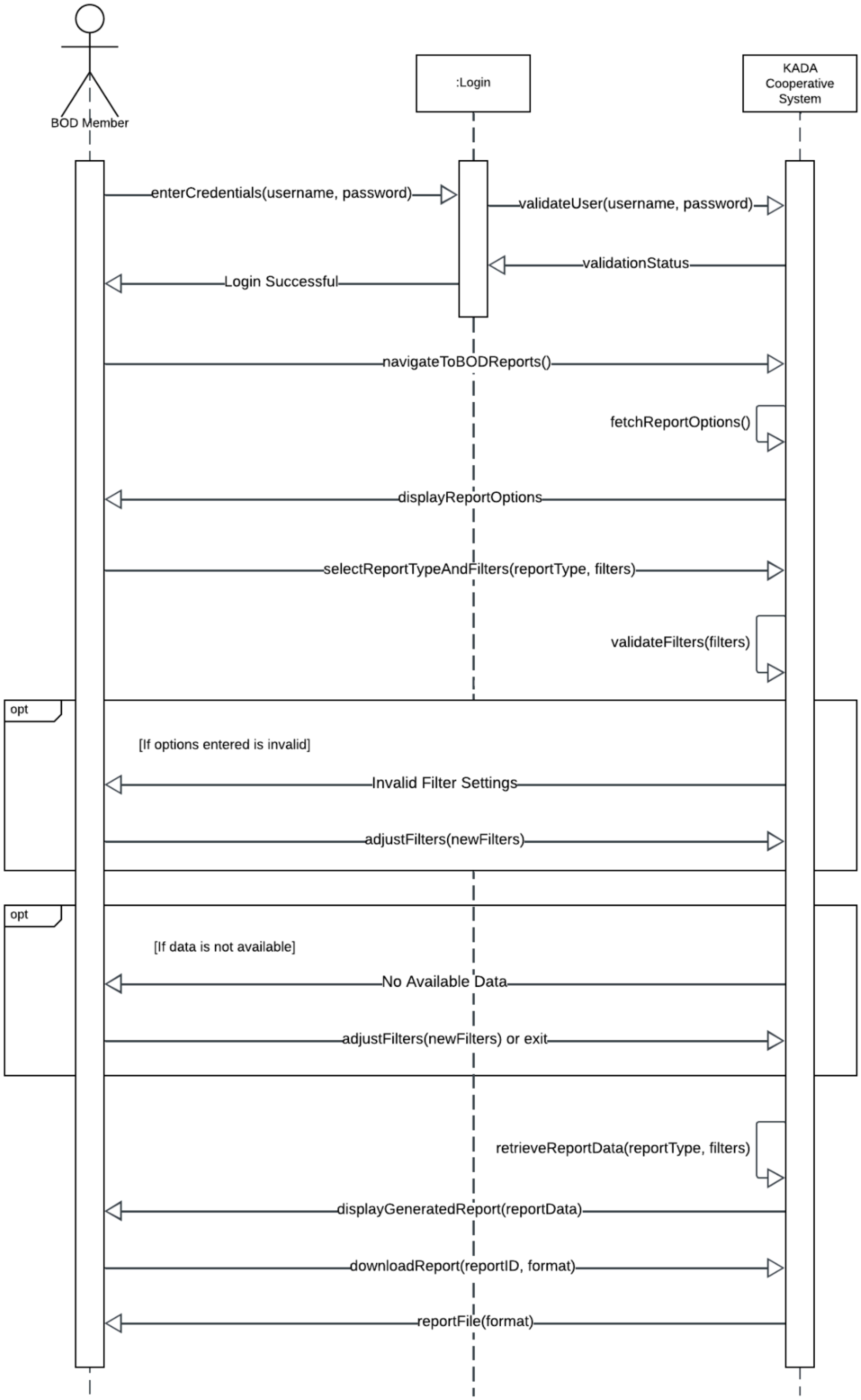


Figure 4.4.7: SD007 Sequence Diagram for <BOD Reports>

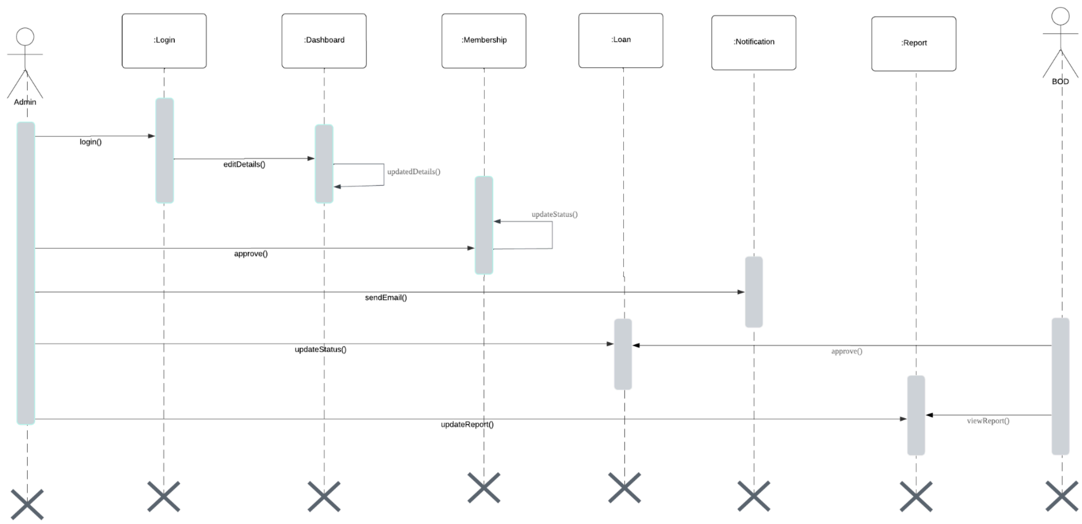


Figure 4.4.8: SD008 Sequence Diagram for <Manage Member>

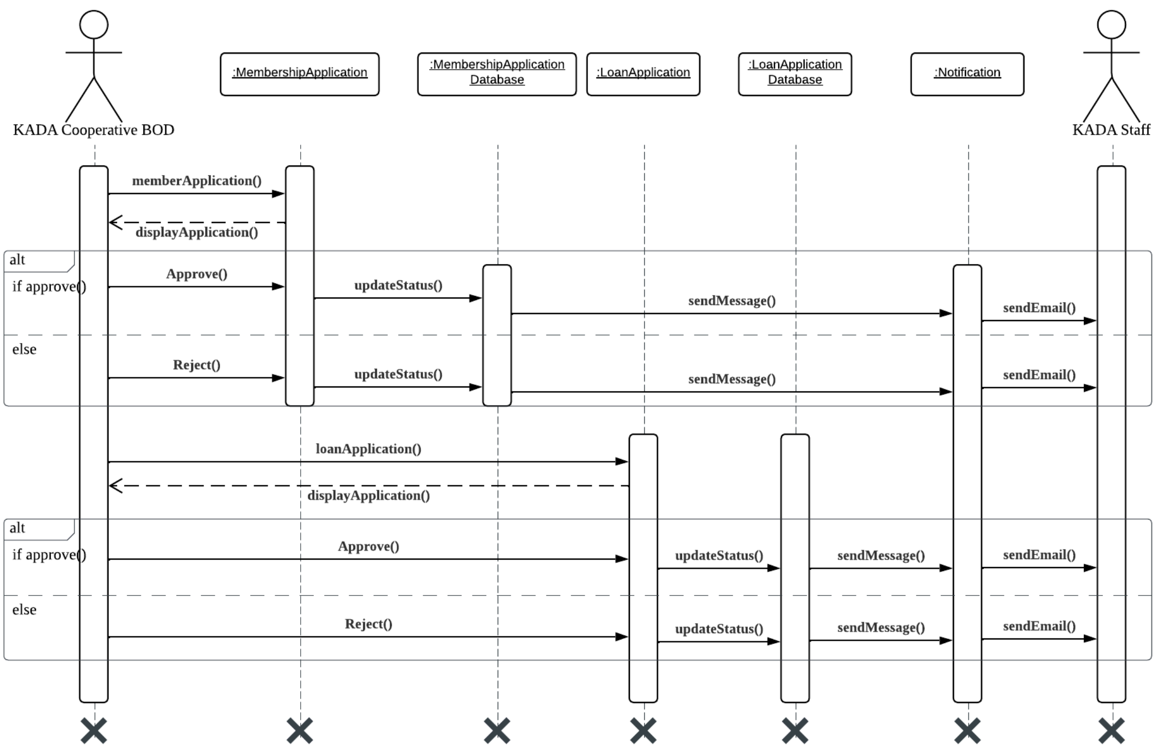


Figure 4.4.9: SD009 Sequence Diagram for <Review Applications>

4.5 **Physical/ Deployment View**

The associated deployment diagram shows the architecture of the KADA system and interaction between its components. The Client (Browser) is a user interface to which a listener is connected securely by Server via HTTPS/TLS. The Server dealing with user requests consists of an Internet Server and an Application Server, manages business logic. To retrieve and save data, it establishes a connection with the Database System (MySQL Database) via DB Connection. This configuration guarantees efficient communication, modularity along with a clear separation of duties so that the whole system can also be scaled properly for management.

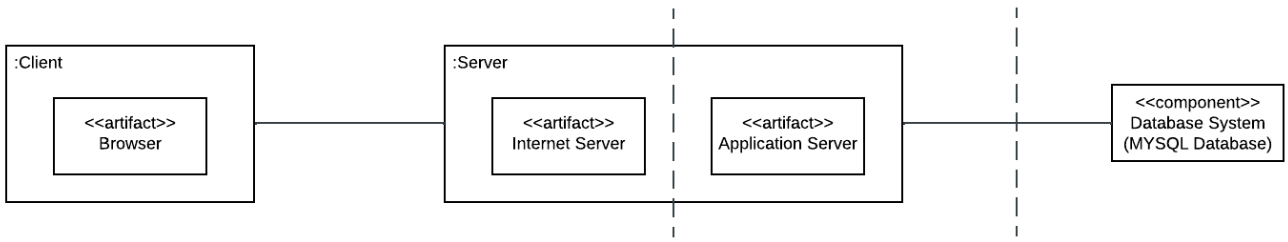


Figure 4.16: Deployment diagram for Internet-based System

5. Data Design

The major data or systems entities are stored into a relational database named as User, BoardOfDirector, KADAAdmin, KADAStaff, NonMember, Member, Membership, FamilyInfo, Loan, Guarantor, Notifications and Stock processed and organized into 12 entities as listed in Table 5.1.

Table 5.1: Description of Entities in the Database

No.	Entity Name	Description
1.	BoardOfDirector	This entity will store the data for the board of directors.
2.	FamilyInfo	This entity will hold the data of the family information for the membership applications.
3.	Guarantor	This entity will hold the data of the guarantors for the loan applications.
4.	KADAAdmin	This entity will hold the data for the administrator.
5.	KADAStaff	This entity will hold the data for the KADA staff.
6.	Loan	This entity will hold the data of the loan applications.
7.	Member	This entity will hold the data for the cooperative member.
8.	Membership	This entity will hold the data of the membership applications.
9.	NonMember	This entity will hold the data for the non cooperative member.
10.	Notifications	This entity will hold the data for the notifications sent to the KADA staff.
11.	Stock	This entity will hold the data of the stock for the cooperative member.
12.	User	This entity will store the data for login processes.

5.1 Data Dictionary

5.1.1 Entity: FamilyInfo

Table 5.1.1: Description of Entity: FamilyInfo

Attribute Name	Type	Description
FamilyMemberIc	INT	Staff's family member Identification card.
FamilyMemberName	VARCHAR	Staff's family member name.
FamilyMemberBirthCerNo	VARCHAR	Staff's family member birth certificate numbers.
FamilyMemberRelationship	TEXT	Family member relationship with KADA members.

5.1.2 Entity: Guarantor

Table 5.1.2: Description of Entity: Guarantor

Attribute Name	Type	Description
guarantorIC	INT	Loan guarantor's identification cards.
guarantorName	VARCHAR	Loan guarantor's name.
guarantorPFNo	INT	Loan guarantor's PF no.
guarantorMemberNo	INT	Loan guarantor's member no.
guarantorPhoneNo	INT	Loan guarantor's phone no.

5.1.3 Entity: KADAStaff

Table 5.1.3: Description of Entity: KADAStaff

Attribute Name	Type	Description
StaffNo	INT	KADA Staff number
ApplicationID	INT	Membership Application ID number.

5.1.4 Entity: Loan

Table 5.1.4: Description of Entity: Loan

Attribute Name	Type	Description
loanApplicationID	INT	Loan application ID
StaffNo	INT	Staff Number / Member No
loanType	TEXT	Loan Type
loanAmount	DOUBLE	Loan Amount
loanPeriod	INT	Loan period in month
monthlyInstallment	DOUBLE	Loan paybank value monthly.

DOB	VARCHAR	Date of birth
age	INT	Staff Age
bankName	TEXT	Bank Name
bankAccountNo	INT	Bank Number Account
loanApplicationStatus	BOOLEAN	Loan application status

5.1.5 **Entity: Membership**

Table 5.1.5: Description of Entity: Membership

Attribute Name	Type	Description
applicationID	INT	Member application ID
email	VARCHAR	KADA non-member’s email
Name	TEXT	Applicants name
icNo	INT	Applicant’s IC No.
MarriageStatus	TEXT	Marriage Status of the applicant
homeAddress	VARCHAR	Applicant’s home address
homePostcode	INT	Applicant home postcode
homeState	TEXT	Applicant’s home state
gender	TEXT	Applicant’s Gender
religion	TEXT	Applicant’s Religion
race	TEXT	Applicant’s race
monthlySalary	DOUBLE	Applicant’s monthly salary
pfNo	INT	Applicant’s pf number
position	TEXT	Applicant’s position
positionGrade	VARCHAR	Applicant’s position grade
workAddress	VARCHAR	Applicant’s work address
workPostcode	INT	Applicant’s work address postcode
faxNo	INT	Applicant’s fax number
phoneNo	INT	Applicant’s phone number
homePhoneNo	INT	Applicant’s home phone number
entranceFee	DOUBLE	Entrance fee
shareCapital	DOUBLE	Share capital
memberDeposit	DOUBLE	Member deposit
charityFund	DOUBLE	Charity fund
fixedSavings	DOUBLE	Fixed savings

applicationDate	DATE	Date of the submission
applicationStatus	TEXT	Membership Application Status

5.1.6 **Entity: Notifications**

Table 5.1.6: Description of Entity: Notifications

Attribute Name	Type	Description
NotificationID	INT	Notification ID
Message	TEXT	Message for the email
Date	DATE	Date of the email
Email	VARCHAR	Email of the recipient

5.1.7 **Entity: Stock**

Table 5.1.7: Description of Entity: Stock

Attribute Name	Type	Description
stockId	INT	Stock ID for each KADA staff member
ShareCapital	DOUBLE	Share capital value
feeCapital	DOUBLE	Fee capital value
fixedSavings	DOUBLE	Total fixed savings balance
membersavings	DOUBLE	Total member savings balance
membersfund	DOUBLE	Total members fund
staffNo	INT	Staff No / Member No

5.1.8 **Entity: User**

Table 5.1.8: Description of Entity: User

Attribute Name	Type	Description
UserID	INT	User ID
email	VARCHAR	Email of each users
password	VARCHAR	Password of each users
role	TEXT	Role of each user.

6. User Interface Design

The interface shows that users can login into the system whether it is the staff, admin or BOD of KADA Cooperative. The interface also provides a homepage tailored to each user role and displaying the relevant features. It also enables the non-members to register as members in KADA Cooperative and for members, they can apply for loans. Users also can view the statement, manage the dashboard and report.

6.1 Screen Images

Here is the detailed user interface design for each use case, showcasing the layout and functionality specific to the tasks and roles involved.

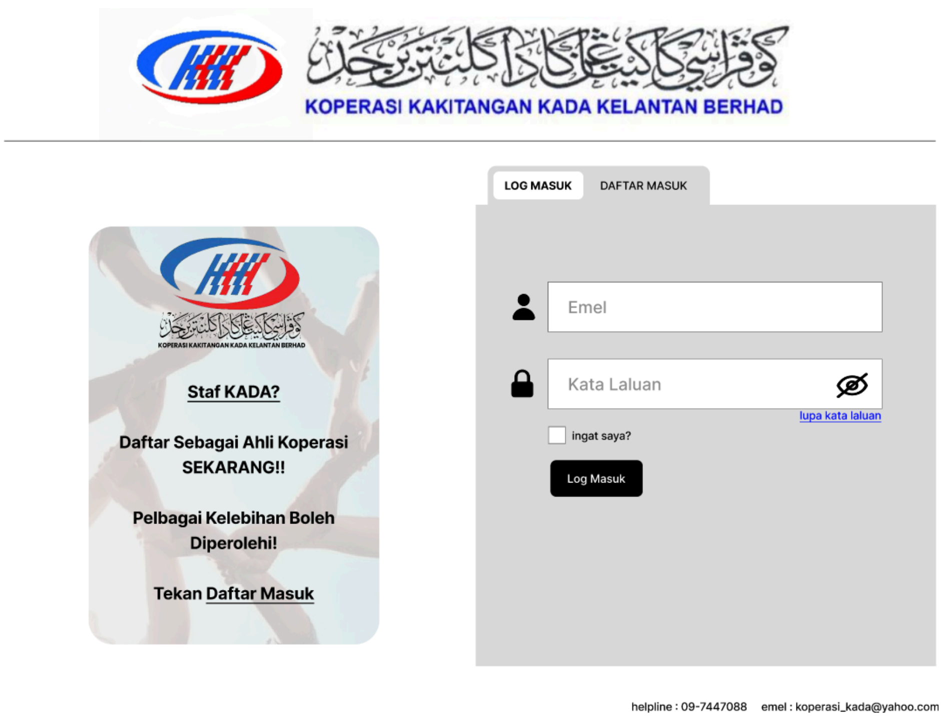


Figure 6.1.1: Interface for Authentication

Figure 6.1.2: Interface for Membership Registration

Figure 6.1.3: Interface for Loan Application

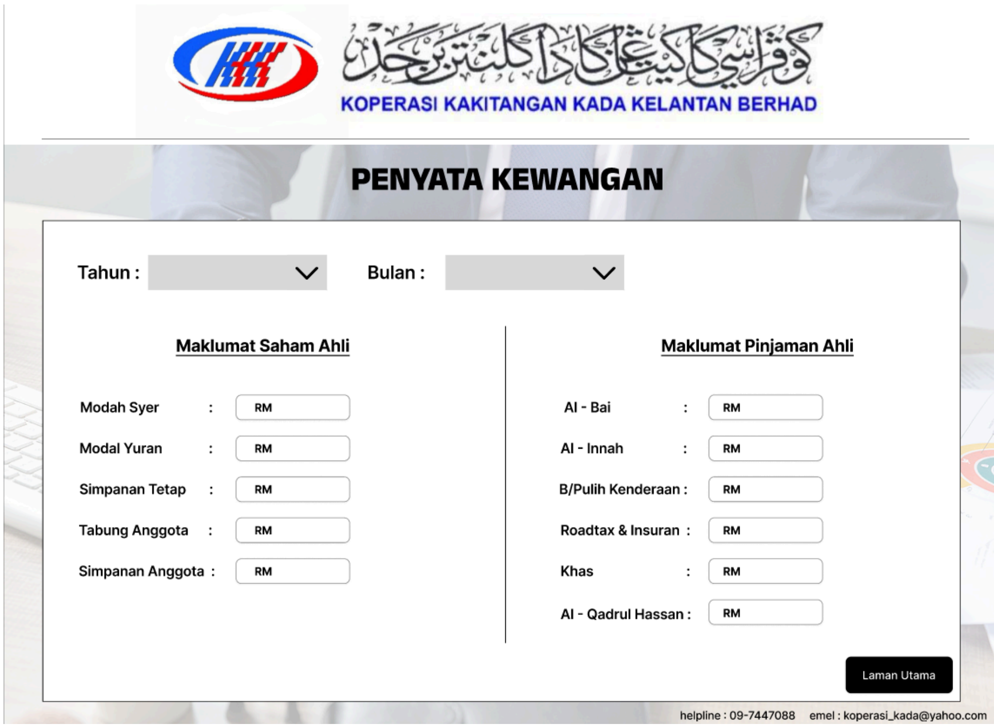


Figure 6.1.4: Interface for View Financial Statement

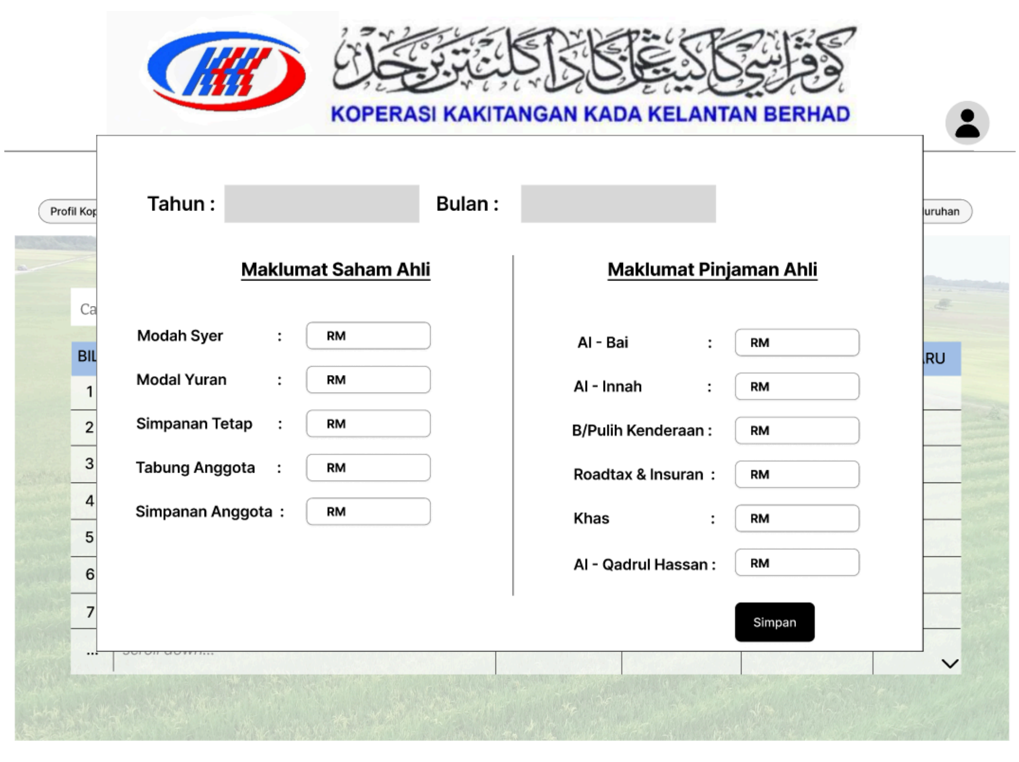


Figure 6.1.5: Interface for Member Statement

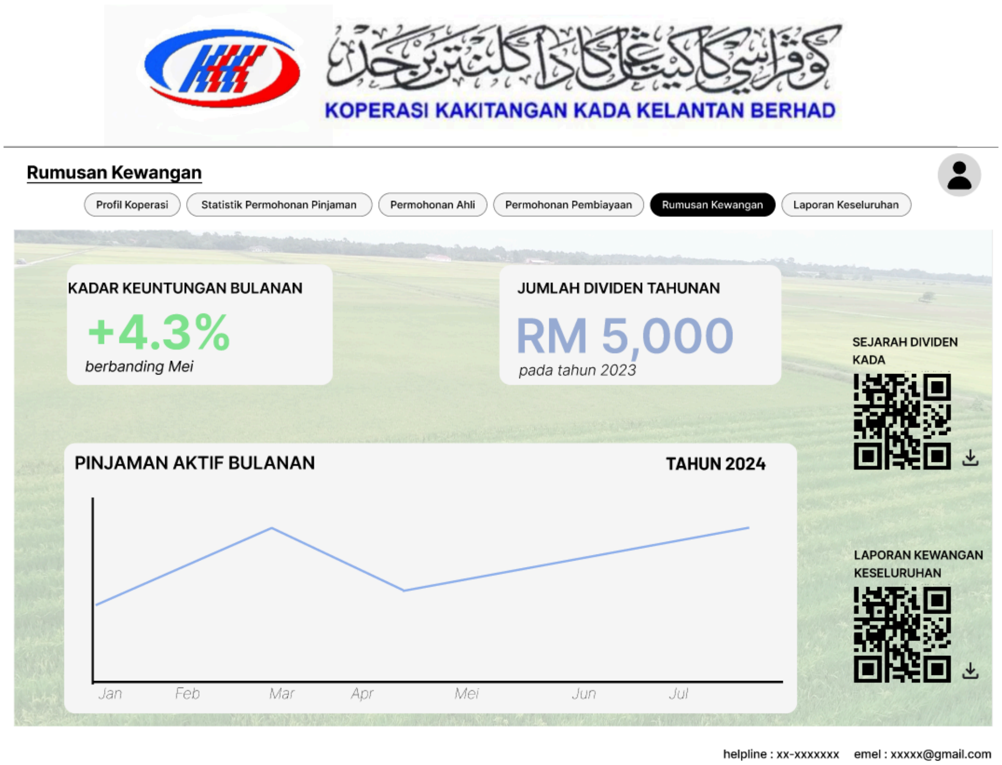


Figure 6.1.6: Interface for View Financial Report

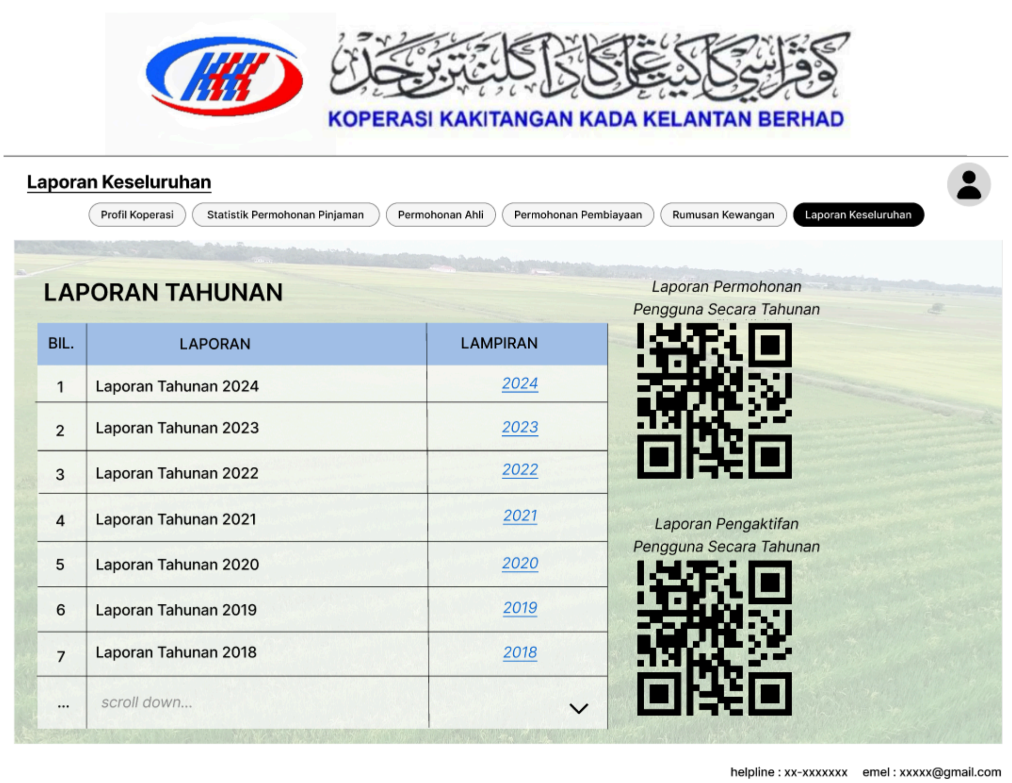


Figure 6.1.7: Interface for BOD Report

Figure 6.1.8: Interface for Manage Member

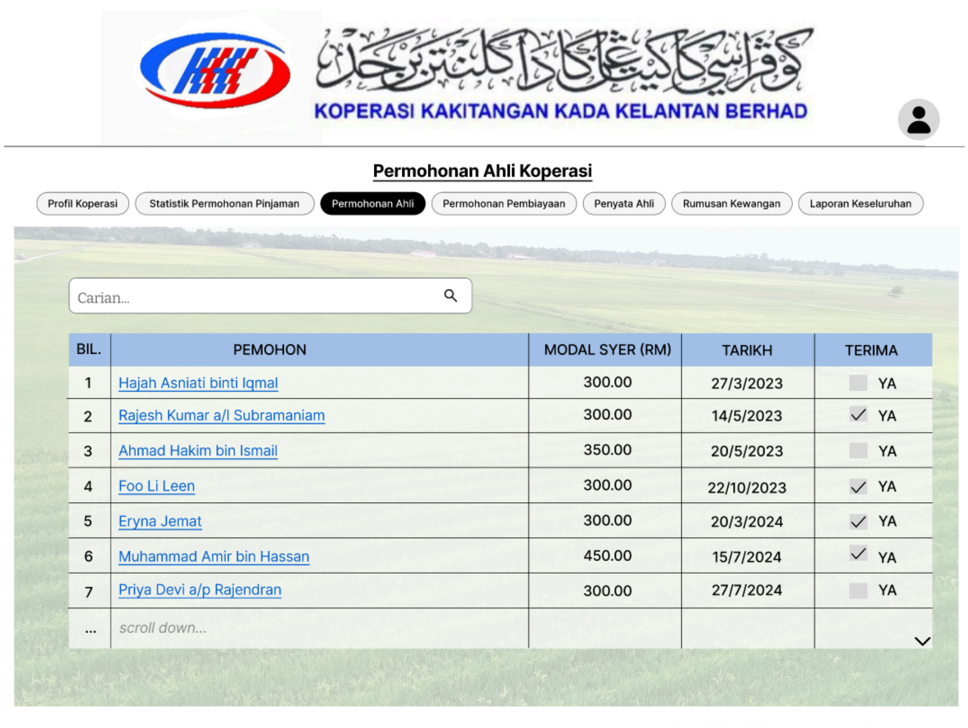


Figure 6.1.9: Interface for Review Application

7. Traceability

This section shows the cross-references to trace the components and the data structures to the requirements of the KADA Cooperative System. The table below shows how each of the system components will satisfy the requirements that are represented as use cases.

Table 7.1: Requirement Matrix

Package, UC , Sequence Diagram /EntityClass	KADAStaff	KADA Admin	BoardOfDirector	Membership	FamilyInfo	Loan	Notifications	Guarantor	Stock
P001, UC001, SD001	X	X	X						
P002, UC001, SD002	X		X	X	X				
P003, UC003, SD003	X		X			X		X	
P004, UC004, SD004	X					X			
P004, UC005, SD005	X					X			
P005, UC006, SD006			X			X			X
P005, UC007, SD007			X						
P006, UC008, SD008		X		X	X				X
P007, UC009, SD009		X	X	X		X		X	X